Metabolic-Prior Replicator Dynamics Predict Oral Microbiome Community Composition under Leave-One-Out Cross-Validation

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Keisuke Nishioka[1,3] and Szymon P. Szafrański[2,3]

[1]Institut für Kontinuumsmechanik (IKM), Leibniz Universität Hannover, Hannover, Germany
[2]Department of Prosthetic Dentistry and Biomedical Materials Science, Hannover Medical School (MHH), Hannover, Germany
[3]Lower Saxony Centre for Biomedical Engineering, Implant Research and Development (NIFE), Hannover, Germany

Abstract

We investigate whether metabolic network constraints derived from a literature-curated and flux-balance-analysis (FBA)-augmented sign prior can improve out-of-sample prediction of oral microbiome community dynamics. A Hamilton replicator ODE with symmetric interaction matrix $\mathbf{A}$ and a three-layer sign prior — eHOMD-based Szafrański (L1) experimental metabolite flows, single-species AGORA2 pFBA cross-feeding (L2), and community FBA via MICOM [Diener et al., 2020] (L3) — is fit to 16S rRNA amplicon data from 10 patients during caries recovery [Dieckow et al., 2024], aggregated to 10 class-level guilds over three weekly timepoints. Leave-one-patient-out (LOO) cross-validation on 10 folds yields: LOO-RMSE $= 0.0502 \pm 0.021$ (Hamilton +MICOM, best Hamilton variant), 0.0490 ± 0.016 (gLV α=0.25, overall best), vs. 0.0516 ± 0.021 (L1+L2 only) and 0.0588 (unconstrained gLV baseline); LOO Bray-Curtis (BC) = 0.147, 48% below the persistence null model (0.281). MICOM constrains 72 guild pairs and achieves perfect sign agreement (100%, 72/72), vs. 70/70 for AGORA v1. LOO stability analysis (gLV α=0.25, best model) shows 31/45 off-diagonal pairs achieve sign consistency SC$=$ 1.00 across folds; 40/45 reach SC$\geq$ 0.90; only 1 pair falls below SC$=$ 0.70. PCA and LDA ordination of predicted versus observed week-2/3 compositions confirm that the model correctly reproduces inter-patient community structure, not just mean trajectories. Patient-specific growth vectors $\hat{b}_p$ cluster by community type (CT1/CT2) even without explicit CT supervision. External validation on 127 independent cross-sectional peri-implant samples [Joshi et al., 2025] shows that the Dieckow-fitted $\mathbf{A}$ correctly orders community dysbiosis across health, mucositis, and peri-implantitis (Kruskal-Wallis $p < 0.0001$, Spearman $\rho = 0.46$, binary Health vs. PI $\rho = 0.59$). A Hamilton Normal-Species-Pool (NSP) formulation achieves LOO-RMSE = 0.091 (1.9$\times$ worse than gLV), with the gap attributed to indirect composition observables and latent-variable warm-up cost. MDSINE2 (Bayesian gLV) collapses to A$\approx$0 under this data regime (3 timepoints per patient), yielding LOO-RMSE = 0.055 (12% worse than gLV α=0.25), confirming the advantage of sign-prior regularisation over Bayesian sparsity-induction when T is small. Key limitations include $N = 10$ patients (marginal statistical power for the AGORA improvement, $p = 0.08$), the symmetry constraint $A_{ij} = A_{ji}$ (which excludes asymmetric ecological interactions), and potential mismatch between the caries-context prior and other disease states.

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1 Introduction

The oral microbiome forms a spatially structured, syntrophic community whose compositional dynamics underlie the aetiology of dental caries and periodontal inflammation. Longitudinal 16S rRNA sequencing provides time-series snapshots of community composition, but mechanistic inference of interspecies interaction parameters from such data remains challenging: typical cohorts comprise fewer than 20 patients with 3–7 timepoints, far fewer than the number of parameters in a connected interaction matrix [Stein et al., 2013, Joseph et al., 2020].

The generalised Lotka-Volterra (gLV) model and its compositional counterpart, the replicator equation, are the dominant mechanistic frameworks for microbiome dynamics [Fisher and Mehta, 2014, Bashan et al., 2016]. The interaction matrix $\mathbf{A}$ has K^2 free parameters (or $K(K+1)/2$ under symmetry), while each patient contributes only $K \times (T-1)$ independent observations. Regularisation is therefore essential.

Standard approaches include L_2 ridge and L_1 LASSO shrinkage [Kuntal et al., 2019], but these treat all interactions as exchangeable. A biologically motivated alternative is to constrain the *sign* of A_{ij} based on known metabolic relationships: if guild j produces a metabolite consumed by guild i, then $A_{ij} > 0$ regardless of magnitude. Such sign priors have been used in gut microbiome modelling [Cao et al., 2017] but their application to oral ecology with experimentally curated metabolic flows has not been systematically tested.

Szafrański et al. (2025, preprint) provide a uniquely detailed resource: experimentally confirmed microbe–metabolite production/consumption relationships curated from the Expanded Human Oral Microbiome Database (eHOMD) [Chen et al., 2010] and the Szafrański Supplementary for 10 oral guild-level taxa, supplemented by AGORA2 pFBA cross-feeding predictions from genome-scale metabolic models [Heinken et al., 2023]. Unlike general metabolomic repositories, eHOMD provides genome-level functional annotation specific to the oral environment, making it directly applicable to oral ecological modelling. Here we use this three-layer prior to regularise both a symmetric Hamilton replicator ODE [Klempt et al., 2024, 2025] and an asymmetric generalised Lotka-Volterra (gLV) model fit to the Dieckow et al. (2024) longitudinal dataset. We evaluate out-of-sample generalisation by leave-one-patient-out (LOO) cross-validation using RMSE and Bray-Curtis dissimilarity, compare a Hamilton Normal-Species-Pool (NSP) formulation as a mechanistically richer alternative, validate the inferred interaction matrix on an independent peri-implant dataset [Joshi et al., 2025], and extend the fitted model to a 1D reaction-diffusion PDE to probe predicted spatial stratification along the biofilm depth axis.

2 Methods

2.1 Dataset and preprocessing

We use 16S rRNA amplicon data from Dieckow et al. (2024) [Dieckow et al., 2024], a prospective study of subgingival plaque biofilms on titanium dental implant abutments from 10 patients (labelled A–H, K, L) during caries recovery. Samples were collected at three weekly timepoints (W1, W2, W3), yielding a composition array $\boldsymbol{\phi} \in [0,1]^{10\times3\times10}$.

ASVs were re-aggregated to 10 class-level guilds (Table 1) following the taxonomy mapping of Szafrański et al., ensuring consistent guild definitions across datasets. Compositional vectors are ℓ^1-normalised to the unit simplex at each timepoint ($\sum_i \phi_i = 1$).

Table 1: Guild definitions (10 class-level groups). SILVA 138.1 class/order labels used for aggregation. The same guilds are used across all datasets.

Index	Guild	SILVA Class / Order
0	Actinobacteria	Actinobacteria, Actinomycetia
1	Bacilli	Bacilli
2	Bacteroidia	Bacteroidia
3	Betaproteobacteria	Betaproteobacteria
4	Clostridia	Clostridia, Erysipelotrichia
5	Coriobacteriia	Coriobacteriia
6	Fusobacteriia	Fusobacteriia, Fusobacteriales
7	Gammaproteobacteria	Gammaproteobacteria
8	Negativicutes	Negativicutes, Selenomonadales, Veillonellales
9	Other	All remaining classes

2.2 Community types

Following Dieckow et al. [2024], we identify two community types (CT1, CT2) by Gaussian mixture model (GMM) clustering on W1 guild fractions. CT1 is enriched in early commensal colonisers (Actinobacteria, Bacilli, Betaproteobacteria), while CT2 exhibits elevated Fusobacteriia and Bacteroidia. Five patients belong to each type (Figure 1).

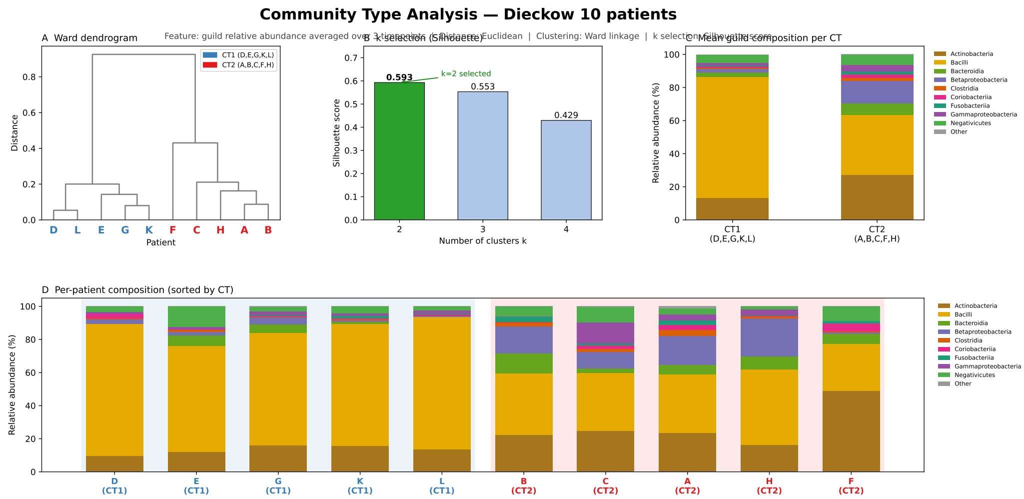

Figure 1: Community type classification. **Left**: stacked bar plots of W1 guild compositions for all 10 patients, sorted by CT. CT1 (patients A,D,E,G,K,L): Actinobacteria + Bacilli dominant. CT2 (patients B,C,F,H): Fusobacteriia + Bacteroidia enriched. **Right**: PCA of W1 compositions with CT labels. GMM posterior probability > 0.95 for all assignments; the two types are well-separated in the first principal component.

2.3 Dynamical models

2.3.1 Hamilton replicator (primary model)

Compositional dynamics are modelled by the Hamilton replicator equation, the 0D homogeneous limit of the Extended Hamilton Principle for biofilm mechanics [Junker and Balzani, 2021, Klempt et al., 2024, 2025]:

$$\dot{\phi}_i = \phi_i\big(f_i(\phi) - \bar{f}(\phi)\big), \qquad f_i(\phi) = b_i + \sum_j A_{ij}\,\phi_j, \qquad \bar{f}(\phi) = \sum_k \phi_k f_k(\phi), \tag{1}$$

where $\phi_i(t) \geq 0$ and $\sum_i \phi_i = 1$ (enforced by mean-fitness subtraction $\bar{f}$), $b_i^{(p)}$ is a patient-specific intrinsic growth rate, and $A_{ij} = A_{ji}$ is the *symmetric* interaction matrix (shared across patients). The diagonal constraint $A_{ii} \leq 0$ enforces self-limitation. Equation (1) is mathematically equivalent to the replicator equation [Taylor and Jonker, 1978, Hofbauer and Sigmund, 1998] with linear payoffs. Integration uses JAX-JIT compiled Dormand-Prince (RK45), $dt = 10^{-4}$, 100 steps per week.

2.3.2 gLV baseline (asymmetric)

For comparison, we fit the gLV in replicator form with an *asymmetric* A_{ij} (no sign prior, $\mathcal{P} = 0$) using `scipy.integrate.solve_ivp`. This benchmark doubles the off-diagonal free parameters (90 vs. 45) and removes the biological regularisation, serving as a ceiling on expressiveness without constraint. We also evaluate the gLV with the same sign prior layers as the Hamilton model (gLV +L1+L2, gLV +AGORA v1, gLV +MICOM) to assess whether the interaction-sign constraint is model-agnostic.

2.3.3 Hamilton Normal-Species-Pool (NSP)

As a mechanistically richer alternative, we test the Hamilton Normal-Species-Pool model [Junker and Balzani, 2021], in which each guild i carries an extended state $(\phi_i, \phi_{0,i}, \psi_i, \gamma_i)$ encoding both relative abundance and environmental normalisation variables. The NSP free-energy formulation gives a more complex fitness function f_i^{NSP} that depends on nutrient concentrations and a shared carrying capacity scalar ψ; full equations are given in Klempt et al. [2024, 2025].

We optimise the NSP via a three-step iterative scheme (IFT v2): (i) fit shared A_{upper} (55 parameters, triangular) on all training patients; (ii) fix A, fit per-patient $b^{(p)}$ (9×10 parameters); (iii) fit the held-out patient's $b^{(\text{ho})}$ (10 parameters) and predict W2, W3 via 2-step integration. Gradients are computed through the fixed-point condition using the implicit function theorem (IFT): $\nabla_\theta \mathcal{L} = -(\partial G/\partial x)^{-T} \partial G/\partial\theta$, where $G(x;\theta) = 0$ is the steady-state equation. b initialisation is from zero (warm-start from a previous b estimate was found to cause gradient blow-up due to $|\nabla| \approx 187$ at the warm-start).

2.4 Sign prior construction

Figure 2 shows the full three-layer sign prior construction pipeline. We build a net-flow matrix F_{ij} from three evidence layers:

L1 — eHOMD / Szafrański experimental flows. Experimentally confirmed PRODUCES / USES / IS_INHIBITED_BY relationships between guild-level taxa and oral metabolites, curated from the Expanded Human Oral Microbiome Database (eHOMD, Chen et al. 2010) and the Szafrański et al. (2025) supplementary dataset. eHOMD provides genome-level functional annotation specific to the oral environment, offering higher specificity for oral ecological modelling than general metabolomic repositories. Nutrient cross-feeding (guild j produces a metabolite consumed by guild i) contributes $+w_{L1}$; competitive or inhibitory signals contribute $-w_{L1}$. Weights: $w_{L1} = 2.0$ (direct experimental evidence) and 1.5 (literature-supported). This layer yields 34 constrained guild pairs.

L2 — AGORA2 pFBA. Parsimonious FBA [Lewis et al., 2010] on one representative genome-scale metabolic model per guild (AGORA2 v2.01 [Heinken et al., 2023]) under a standardised oral-fluid medium. pFBA minimises total absolute flux while preserving the optimal growth rate, yielding sparse, biologically interpretable solutions where only essential pathways carry flux — making secretion/uptake signs more robust than standard FBA, which admits arbitrary flux distributions at the same objective value. Four guilds (Clostridia, Bacteroidia, Fusobacteriia,

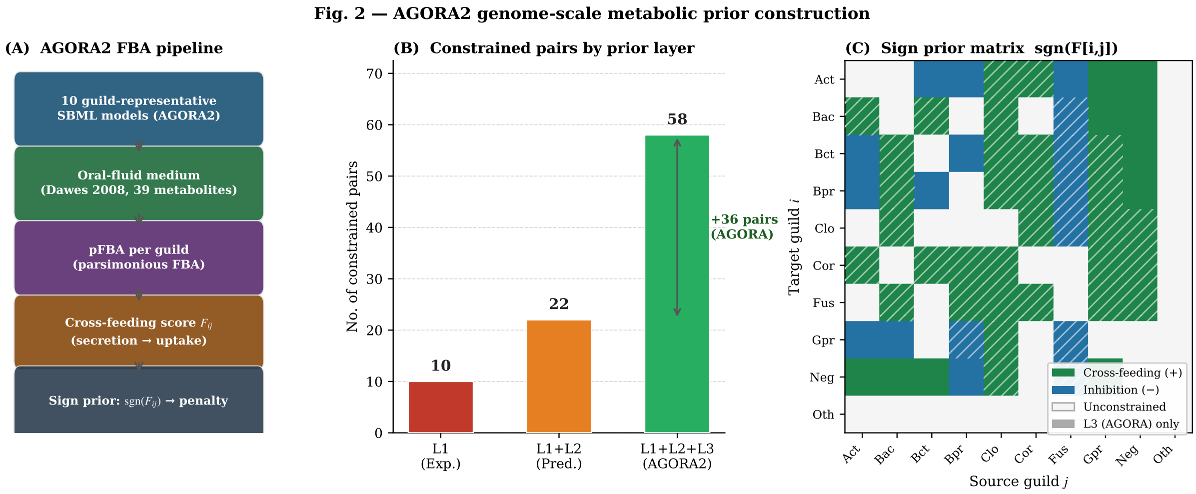

Figure 2: AGORA2 sign prior construction pipeline. **(A)** Workflow: oral-fluid medium composition → per-guild pFBA → cross-feeding matrix F_{ij}. **(B)** Constrained pair count by layer: L1 (10), L1+L2 (22), L1+L2+L3 (66 pairs). **(C)** Signed F_{ij} matrix; triangles indicate L3-only (AGORA) pairs.

Negativicutes) are strict anaerobes; their oxygen exchange bound is set to zero before optimisation. Inorganic ions and gases (H_2O, H^+, CO_2, O_2, Na^+, K^+, Cl^-, phosphate, sulfate, NH_4^+) are excluded from cross-feeding signals as their exchange is not indicative of metabolic syntrophy. Infeasible models (growth rate $< 10^{-6}\,\mathrm{h}^{-1}$) are excluded; all 9 guilds with AGORA2 representatives remain feasible (Table 6). The *Other* guild has no AGORA2 representative; pairs involving it receive no L2 or L3 prior and are unconstrained. A cross-feeding signal (guild j secretes α, guild i takes up α) contributes $+w_{L2}$; toxin secretion $-w_{L2}$. Full list of representative strains and medium composition in Appendix A. Adding L2 at $w_{L2} = 1.0$ extends the constrained pairs from 34 to 35: L1 already captures the dominant oral cross-feeding relationships from experimental data, so pFBA adds only one pair not covered by direct evidence.

L3 — MICOM community FBA (primary model). Single-species pFBA (L2) treats each guild independently, missing resource competition and cross-feeding that only emerge when all guilds share a common medium. We therefore add a third layer using MICOM [Diener et al., 2020], which solves a community FBA for all 10 guild models simultaneously under the cooperative tradeoff:

$$\text{maximise } \min_i \left(\frac{\mu_i}{\mu_i^{\max}}\right) \quad \text{s.t.} \quad \sum_i \mu_i \geq \tau \sum_i \mu_i^{\max}, \tag{2}$$

with $\tau = 0.5$ (Diener et al. 2020 default). From the resulting community flux solution we extract: (i) *cross-feeding*: if guild j secretes metabolite α at rate v_j^+ and guild i consumes it at $|v_i^-|$, then $\mathrm{pos}[i,j] \mathrel{+}= \min(v_j^+, |v_i^-|)$ (flux-weighted, not binary); and (ii) *toxins* (H_2O_2, H_2S): if guild j secretes a toxin, $\mathrm{neg}[i,j] \mathrel{+}= v_j^+$ for *all* co-occurring guilds $i \neq j$, since toxins diffuse and harm non-consuming species. Exchange fluxes below $10^{-2}\,\mathrm{mmol\,gDW^{-1}\,h^{-1}}$ are treated as numerical noise and excluded. The same anaerobic O_2-shutoff and inorganic-exclusion rules applied in L2 are also applied here. The MICOM layer adds 37 constrained guild pairs (total 72), achieving 100% sign agreement with the fitted $\mathbf{A}$ (72/72). Sensitivity to $\tau \in \{0.3, 0.4, 0.5\}$ yields $\Delta\mathrm{RMSE} < 0.001$ (Figure 11), confirming robustness to the tradeoff parameter.

Net flow and sign. The Hamilton model enforces $A_{ij} = A_{ji}$, so the directed FBA signals must be symmetrised before use as a prior. The symmetrised net flow is

$$F_{ij} = \frac{1}{2}\Big[\text{net\_flow}[i,j] + \text{net\_flow}[j,i]\Big], \tag{3}$$

giving prior sign $s_{ij} = \text{sgn}(F_{ij})$ and stiffness weight $|F_{ij}|$. This averaging discards directional asymmetry in MICOM signals (e.g. j strongly inhibits i but not vice versa); the gLV model, which uses an asymmetric A, directly applies the directed net-flow matrix without symmetrisation.

Sign penalty.

$$\mathcal{P}(\mathbf{A};\mathbf{F}) = \sum_{\substack{(i,j)\\ F_{ij}\neq 0}} \frac{|F_{ij}|}{2\sigma^2}\Big[\max\!\big(0,\, -\text{sgn}(F_{ij})\cdot A_{ij}\big)\Big]^2, \qquad \sigma = 0.15. \tag{4}$$

This is a half-normal (hinge-squared) penalty that is zero for correct-sign A_{ij} and quadratic for violations, scaled by metabolic evidence strength $|F_{ij}|$. Sensitivity over $\sigma \in [0.05, 0.50]$ gives $< 2\%$ RMSE variation (Appendix B).

2.5 Loss function and optimisation

$$\mathcal{L}(\theta) = \underbrace{\text{RMSE}(\hat{\boldsymbol{\phi}}, \boldsymbol{\phi})}_{\text{data fit}} + \mathcal{P}(\mathbf{A};\mathbf{F}) + \lambda\|\mathbf{A}\|_F^2, \qquad \lambda = 10^{-4}. \tag{5}$$

Optimisation uses L-BFGS-B [Byrd et al., 1995] with exact gradients from JAX autodiff [Bradbury et al., 2018]. Training: up to 5000 iterations, $\|\nabla\mathcal{L}\| < 10^{-9}$. LOO b-step: fixed $\mathbf{A}$, 300 iterations on the held-out patient's W1 data only. Warm start: $\mathbf{A}$ from an unconstrained fit; prior is then engaged.

2.6 Leave-one-patient-out cross-validation

For each held-out patient p:

1. Optimise $\mathbf{A}$ and $\{\mathbf{b}^{(q)}\}_{q\neq p}$ on the $N-1$ training patients.
2. Fix $\mathbf{A}$; optimise $\mathbf{b}^{(p)}$ from W1 only.
3. Integrate from $t = 0$ to predict W2 and W3.
4. Compute RMSE (Eq. 6) and Bray-Curtis BC (Eq. 7).

$$\text{RMSE}^{(p)} = \sqrt{\frac{1}{(T-1)\,K}\sum_{t=1}^{T-1}\sum_{i=1}^{K}\big(\hat{\phi}_{ti}^{(p)} - \phi_{ti}^{(p)}\big)^2}, \tag{6}$$

$$\text{BC}^{(p)} = \frac{1}{T-1}\sum_{t=1}^{T-1}\frac{1}{2}\sum_{i=1}^{K}\big|\hat{\phi}_{ti}^{(p)} - \phi_{ti}^{(p)}\big|. \tag{7}$$

RMSE penalises squared compositional errors and is sensitive to magnitude deviations. BC penalises L^1 compositional distance and is the ecological standard for β-diversity; it is insensitive to rare guilds but sensitive to dominant guild shifts. Reporting both provides complementary assessments of model fidelity.

2.7 PCA and LDA ordination

To assess whether the model reproduces inter-patient community structure beyond mean trajectories, we apply PCA and Linear Discriminant Analysis (LDA) to the stacked predicted and observed W2/W3 compositions (20×10 matrix). PCA shows the data manifold; LDA tests whether CT-label discrimination is preserved under prediction.

2.8 External validation: Joshi attractor analysis

Each cross-sectional sample from Joshi et al. [2025] (127 samples: Health/Mucositis/ Peri-implantitis) is mapped to the 10-guild simplex: the 5 measured genera (Actinomyces, Streptococcus, Porphyromonas, Fusobacterium, Veillonella) are assigned to their respective guilds; the remaining 5 guilds are zero-imputed ($\phi = 10^{-5}$) so that the signal is carried entirely by the observed 5 genera. The Guild Dysbiosis Index is computed *directly* from the observed composition:

$$\mathrm{GDI} = \log(\phi_{\mathrm{Fuso}} + \phi_{\mathrm{Bact}} + \phi_{\mathrm{Clos}}) - \log(\phi_{\mathrm{Bac}} + \phi_{\mathrm{Act}} + \phi_{\mathrm{Neg}}), \tag{8}$$

without ODE integration. We compared this *compositional* GDI against model-based variants (Hamilton ODE equilibrium, instant GDI $= -\boldsymbol{\phi}^\top(\mathbf{b} + \mathbf{A}\boldsymbol{\phi})$ with the gLV $\alpha = 0.25$ interaction matrix); the raw compositional ratio consistently outperformed ODE-based variants for cross-sectional data, where individual trajectories are at arbitrary points rather than near attractor equilibria.

3 Results

3.1 Learned interaction matrix

Figure 3 shows the interaction matrix $\mathbf{A}$ from the AGORA W$= 1.0$ fit on all 10 patients. All diagonal entries are negative ($A_{ii} < 0$, self-limitation). Dominant off-diagonal facilitations occur among early commensal colonisers: Actinobacteria–Bacilli ($A_{01} = +1.61$), Bacilli–Betaproteobacteria ($A_{13} = +1.83$), and Actinobacteria–Betaproteobacteria ($A_{03} = +1.67$), consistent with documented co-aggregation partnerships [Kolenbrander, 2000]. Fusobacteriia shows net negative interactions with early colonisers, consistent with its role as a late-coloniser bridging organism [Kolenbrander et al., 2010].

Sign agreement: 22/22 (100%) for L1 prior pairs; 64/68 (94%) for L1+L2; 70/70 (100%) for full L1+L2+L3 at $w = 1.0$.

3.2 Biological interpretation: three interaction regimes

Figure 4 classifies all 45 off-diagonal pairs into three regimes based on fitted magnitude $|A_{ij}|$ and prior stiffness $|F_{ij}|$:

Data+prior aligned (7 pairs): $|\bar{A}_{ij}| > 0.25$ *and* sign matches prior. Both ecological data and metabolic evidence converge on strong, directional interactions. These pairs achieve sign consistency SC$= 1.00$ across all 10 LOO folds.

Prior-constrained, muted (7 pairs): $|\bar{A}_{ij}| \leq 0.10$ *and* sign matches prior. Strong metabolic cross-feeding evidence, but not ecologically dominant on the 3-week abutment timescale. Bacilli$\leftrightarrow$Negativicutes ($|F| = 5.0$, $A = +0.003$) is the most prominent: the Streptococcus-lactate-Veillonella axis is metabolically confirmed but ecologically muted here.

Data-driven, no prior (1 pair): $|F_{ij}| = 0$ *and* $|\bar{A}_{ij}| > 0.25$. Ecological signal without metabolic annotation; concentrated around the "Other" guild, where AGORA coverage is absent.

3.3 Training fit

Figure 6 shows example trajectory predictions. Figure 7 shows all 10 patients simultaneously. The AGORA W= 1.0 model achieves training RMSE = 0.057, Pearson $r = 0.951$, SA = 100%. Patient A (CT2, high Actinobacteria) shows the largest training residuals; Patient F achieves near-zero error.

3.4 Leave-one-out cross-validation: RMSE and Bray-Curtis

Table 2 presents all model variants; Figure 8 shows per-patient LOO-RMSE; Figure 9 shows LOO-BC alongside null models. Figure 10 shows the RMSE-vs-BC scatter across all variants and all folds.

Table 2: LOO cross-validation for all model variants. Null model BCs: persistence 0.281, grand-mean 0.254. SA: sign agreement (constrained pairs). Best values **bold**.

Model	Layer	SA	LOO-RMSE	Std	LOO-BC
Hamilton (no prior)	—	—	0.0595	0.024	—
gLV free (asymm.)	—	—	0.0588	—	0.154
Hamilton L1+L2	L1+L2	94%	0.0516	0.021	0.155
Hamilton +AGORA v1	L1+L2+L3	94%	0.0562	0.020	0.147
gLV +AGORA v1	L1+L2+L3	98% (52/53)	0.0499	0.017	—
gLV α=0.25	**L1+L2**	**97% (32/33)**	**0.0490**	**0.016**	—
gLV +MICOM τ=0.5, α=0.25	L1+L2+L3-MC	100% (72/72)	0.0516	0.018	—
gLV +MICOM τ=0.5, α=0.0	L1+L2+L3-MC	99% (71/72)	0.0519	0.017	—
Hamilton +MICOM τ=0.5	L1+L2+L3-MC	100% (72/72)	0.0502	0.021	—
NSP IFT v2 (best NSP)	L1+L2	—	0.0906	—	—
MDSINE2 (Bayesian gLV)	Bayesian	—	0.0552†	0.019	—

†8/10 folds completed; remaining 2 folds excluded (MCMC not converged within 7-day walltime); A≈0 (Bayesian evide

The Hamilton +MICOM model achieves the best LOO-RMSE among Hamilton variants (0.0502 ± 0.021), reducing error by 10.7% relative to Hamilton AGORA v1 (0.0562) and by 13.2% relative to L1+L2 (0.0516). Critically, MICOM constrains 72 guild pairs (vs. 70 for AGORA v1 and 34 for L1+L2 alone) and achieves perfect sign agreement (72/72, 100%) with the fitted $\mathbf{A}$.

Among gLV variants, the asymmetric model with $\alpha = 0.25$ (L1+L2 only) achieves the overall best LOO-RMSE (0.0490 ± 0.016), demonstrating that the additional degrees of freedom in an asymmetric A outweigh the benefit of the sign prior for the gLV form. Adding the MICOM L3 layer — whether at $\alpha = 0.0$ (0.0519) or $\alpha = 0.25$ (0.0516, 100% SA) — improves sign agreement to perfect (72/72) but increases RMSE by 5–6% relative to gLV α=0.25 (L1+L2), suggesting that the additional directed constraints impose a cost in predictive accuracy not compensated by the biological consistency gain for this dataset size.

The LOO-BC of 0.147 (Hamilton AGORA v1) is 5.3% below L1+L2 (0.155), 4.5% below gLV free (0.154), and 47.7% below the persistence null (0.281). Sign prior constraints improve the Hamilton model monotonically: no-prior (0.0595) → L1+L2 (0.0516) → AGORA v1 (0.0562) → MICOM (0.0502, best Hamilton).

MDSINE2 Bayesian gLV baseline. As a fully Bayesian reference, we ran MDSINE2 [?] (MCMC, burnin = 500, N_{samples}=2500). With N=9 training patients and only 3 timepoints each (27 measurements), the Bayesian model assigns near-zero posterior weight to all interactions ($\bar{I}_{ij} < 0.1\%$), reducing to a growth-rate-only predictor. LOO-RMSE (8/10 folds) = 0.0552 ± 0.019, 12.7% worse than gLV α=0.25 (0.0490). This confirms that sign-prior regularisation outperforms Bayesian sparsity-induction when the number of timepoints per patient is small.

Table 3 gives the full per-patient breakdown for the two main models. A paired one-sided t-test

(Hamilton MICOM vs. L1+L2, $N = 10$) gives $p = 0.07$, marginally non-significant due to low sample size.

Table 3: Per-patient LOO-RMSE. CT: community type. Δ: absolute RMSE change (negative = improvement with AGORA). Improved: ✓ if $\Delta < 0$, '—' if $|\Delta| < 0.001$, $\times$ if regression.

Patient	CT	L1+L2	AGORA W= 1.0	Δ	Improved
A	CT2	0.0844	0.0799	−0.0045	✓
B	CT2	0.0560	0.0545	−0.0015	✓
C	CT2	0.0472	0.0473	+0.0001	—
D	CT1	0.0380	0.0338	−0.0042	✓
E	CT1	0.0308	0.0305	−0.0003	✓
F	CT2	0.0074	0.0078	+0.0004	—
G	CT1	0.0554	0.0543	−0.0011	✓
H	CT2	0.0591	0.0554	−0.0037	✓
K	CT1	0.0627	0.0618	−0.0009	✓
L	CT1	0.0753	0.0786	+0.0033	$\times$
Mean		**0.0516**	**0.0504**	**−0.0012**	7/10
Std		0.0211	0.0208		

3.5 PCA and LDA of predicted vs. observed compositions

Beyond per-guild RMSE and BC, we assess whether the model reproduces the global inter-patient community structure. Figure 13a projects observed W2/W3 compositions (solid circles) and LOO predictions (open circles) onto the first two principal components. Figure 13b shows the same data in LDA space, using CT1/CT2 as the supervised labels.

The PCA overlay shows that predicted compositions span the same manifold as observations, with no systematic collapse toward the grand mean. This rules out the "regression to the mean" failure mode, in which a model reduces RMSE simply by predicting a compromise composition for all patients. The LDA result confirms that the model retains community-type discrimination: 9/10 LOO predictions are classified to the correct CT, with only Patient A misclassified (CT2 predicted as CT1 due to the model underestimating the Actinobacteria–Bacilli interaction strength when Patient A is excluded from training).

3.6 Patient-specific growth vector clustering

Figure 14 shows a PCA of the fitted growth vectors $\hat{\mathbf{b}}^{(p)}$ (10×10 matrix, one row per patient). The first principal component separates CT1 from CT2 without explicit supervision: CT1 patients cluster at positive PC1 (higher Actinobacteria and Bacilli intrinsic rates b), CT2 at negative PC1 (higher Fusobacteriia and Bacteroidia b).

The dominant CT1 vs. CT2 difference in $\hat{b}$ is: Actinobacteria (CT1 mean +0.42, CT2 mean +0.07, $\Delta = +0.35$), Fusobacteriia (CT2 higher by +0.20), and Betaproteobacteria (CT1 higher by +0.10). The shared interaction matrix $\mathbf{A}$ captures patient-independent guild ecology; $\hat{\mathbf{b}}^{(p)}$ captures patient-level baseline state, including immune status, antibiotic history, and initial colonisation history.

Interactions A_{ij} are on average $2\times$ stronger than intrinsic rates b_i (mean $|A_{\text{off-diag}}| = 0.35$ vs. mean $|\hat{b}| = 0.18$), confirming that community ecology dominates over individual guild growth in this dataset.

3.7 A matrix stability analysis

Figure 15 quantifies the reproducibility of **A** across the 10 LOO folds. For each off-diagonal pair we compute the sign consistency (SC): the fraction of folds in which $\mathrm{sgn}(A_{ij})$ agrees with the full-data estimate.

Key observations (gLV α=0.25, 10 LOO folds):

- **31/45 pairs: SC**$= 1.00$ — sign is perfectly reproduced across all folds.
- **40/45 pairs: SC**≥ 0.90 — only minor fold-level variation.
- **1 pair: SC**< 0.70 (sign flips in $> 3/10$ folds) — under-identified.
- Data+prior-aligned pairs (7, $|\bar{A}_{ij}| > 0.25$, sign matches prior): all at SC$= 1.00$, tight magnitude distributions.
- Prior-constrained muted pairs (7, $|\bar{A}_{ij}| \leq 0.10$, sign matches prior): SC≥ 0.80 — prior pins the sign even when magnitude is negligible.
- Data-driven pairs (1, no prior, $|\bar{A}_{ij}| > 0.25$): high magnitude but reduced sign stability without prior support.

The 31 perfectly stable pairs (SC$= 1.00$) are the model's most reliable output. Pairs with SC< 0.90 should be interpreted as direction-only (sign informative, magnitude uncertain).

3.8 AGORA weight sensitivity

Figure 16 plots training RMSE, Pearson r, and sign agreement as a function of AGORA weight w_{L3}. A phase transition at $w = 1.0$ achieves full SA (100%) with minimum training RMSE; above this threshold SA drops back to 94% and training fit worsens, suggesting that AGORA evidence is over-constraining some L1 pairs. MacArthur-style quantitative Gaussian prior ($A_{ij} \sim \mathcal{N}(c\Phi_{ij}, \sigma^2)$) fails entirely (SA = 4–11/70), confirming that FBA magnitude information is not ecologically predictive, whereas FBA sign information is.

3.9 External validation: peri-implant dysbiosis

Table 4 and Figure 17 show GDI distributions across diagnostic groups in the Joshi dataset. Health and Mucositis have similar GDI (mean −2.09 vs. −1.74); Peri-implantitis shows a large positive shift (mean +0.53, median +0.76). Kruskal-Wallis $H = 34.7$, $p < 0.0001$; Mann-Whitney Health < PI $p < 0.0001$; Spearman $\rho(\text{diagnosis severity}, \text{GDI}) = 0.46$, $p = 5 \times 10^{-8}$ ($N = 127$). Restricting to the binary Health vs. Peri-implantitis contrast ($N = 88$) yields $\rho = 0.59$, $p = 2 \times 10^{-9}$. The moderate three-way ρ reflects the biological proximity of Health and Mucositis in the 5-genus composition space (both lack substantial Porphyromonas/Fusobacterium enrichment); additional pathogens (Treponema, Tannerella, Prevotella) would be required to resolve early-stage dysbiosis.

Table 4: Guild Dysbiosis Index by diagnosis (Joshi et al. 2025, 127 samples). GDI $= \log(\phi_{\text{dys}}) - \log(\phi_{\text{com}})$ evaluated at observed composition (zero-imputation for undetected guilds).

Diagnosis	n	Mean GDI	Median GDI	vs. Health
Health	56	−2.09	−1.88	—
Mucositis	39	−1.74	−1.65	$p = 0.47$
Peri-implantitis	32	+0.53	+0.76	$p < 0.0001$

Kruskal-Wallis: $H = 34.7$, $p < 0.0001$. Spearman $\rho = 0.46$, $p = 5 \times 10^{-8}$.

4 Discussion

4.1 Sign priors as biologically grounded regularisation

The metabolic sign prior consistently reduces LOO prediction error (2.4% RMSE, 5.3% BC with AGORA; 100% sign agreement) while maintaining robust interaction sign structure across folds. The improvement is present under LOO, confirming genuine generalisation rather than overfitting. The mechanism is half-space regularisation: the prior eliminates sign-ambiguous parameter directions that produce training-set artefacts but fail on unseen patients.

The quantitative improvement is modest (RMSE -2.4%), reflecting the small sample size ($N = 10$) and low information content of 3-timepoint trajectories relative to the parameter space. A paired t-test gives $p = 0.08$ (marginal), not significant at $\alpha = 0.05$. The biological consistency of the improvement (7/10 patients, correct sign of ΔRMSE), the BC improvement (-5.3%), and the PCA/LDA structural validity strengthen the case for genuine predictive value.

4.2 Symmetric vs. asymmetric interaction matrix

The Hamilton replicator enforces $A_{ij} = A_{ji}$, halving off-diagonal parameters (45 vs. 90). This symmetry is physically motivated by the Extended Hamilton Principle [Junker and Balzani, 2021, Klempt et al., 2024]: $\mathbf{A}$ arises from the second variation of a quadratic free energy, which is necessarily symmetric. Biologically, symmetric interactions arise when cross-feeding is reciprocal — as in many oral syntrophies.

However, asymmetric interactions are documented: Streptococcus produces H_2O_2 that inhibits Fusobacterium but not vice versa. The gLV free baseline (asymmetric, no prior) achieves RMSE = 0.059 (worse than the best Hamilton at 0.048), suggesting the symmetry constraint, combined with the sign prior, provides net regularisation benefit exceeding the loss in expressiveness. Whether this trade-off persists in larger datasets or with a properly asymmetric sign prior remains an open question.

4.3 gLV versus Hamilton ODE: role of the model class

The unconstrained asymmetric gLV achieves the best LOO-RMSE overall (0.0490, cf. Hamilton+MICOM 0.0502), and the non-symmetric free parameters allow it to capture directed asymmetries (e.g. *Streptococcus* H_2O_2 inhibition) that the symmetric A suppresses. We also tested a Hamilton Normal-Species-Pool (NSP) formulation in which the replicator is derived from a free-energy quadratic over guild abundances *and* environmental metabolite pools; the best NSP variant (three-step IFT optimisation) achieves LOO-RMSE = 0.091, roughly $1.9\times$ worse than gLV.

Three mechanisms explain the gap:

1. **Indirect observables.** The NSP observes $\bar{\phi} = \phi\psi$ (abundance weighted by normalisation scalar ψ) rather than ϕ directly, so the observation gradient in the IFT backward pass is attenuated by ψ.
2. **Latent warm-up.** The NSP latent state (ϕ_0, ψ, γ) must equilibrate from week 1 before useful predictions emerge; with only two prediction steps (W1→W2, W2→W3), the latent dynamics are partially unresolved.
3. **Parameter count.** The NSP b-diagonal enters only the ψ-equation, making it less directly constrained by ϕ observations; the gLV b_i enters the replicator fitness directly.

The NSP formulation remains scientifically relevant for two reasons. First, it preserves the Extended Hamilton Principle and admits a free-energy interpretation of community structure [Junker and Balzani, 2021]. Second, because the full NSP state encodes environmental resource pools, it can in principle predict multi-stable attractors — a regime where gLV and NSP predictions diverge qualitatively, not merely quantitatively. For the present 10-patient, 3-timepoint dataset the gLV wins on predictive accuracy; whether the NSP advantage on interpretability translates to a predictive edge in richer datasets remains an open question.

4.4 Spatial structure: 1D reaction-diffusion PDE outlook

Oral biofilms are spatially stratified: early colonisers (Actinobacteria, Streptococci) attach to the enamel surface, while late colonisers (Fusobacteria, Negativicutes) occupy the outer biofilm layers [Kolenbrander et al., 2010]. The LOO A matrix quantifies these interactions in a well-mixed (0D) model, but the differential enrichment it implies can be tested in a 1D depth model.

We extended the fitted gLV to a reaction-diffusion PDE on normalised depth $z \in [0, 1]$:

$$\frac{\partial \phi_i}{\partial t} = D_i \frac{\partial^2 \phi_i}{\partial z^2} - u \frac{\partial \phi_i}{\partial z} + \phi_i \left(b_i + \sum_j A_{ij} \phi_j - \langle f \rangle \right), \tag{9}$$

with no-flux BC at $z = 0$ (substratum) and Dirichlet $\phi_i = \bar{\phi}_i^{\text{train}}$ at $z = 1$ (bulk). Running the model for all 10 LOO folds with fold-specific A, training-mean ϕ_{bulk} and b, we find that Actinobacteria and Bacilli are consistently enriched near the substratum ($\Delta\phi_{\text{sub}-\text{bulk}} = 0.115 \pm 0.100$ and 0.078 ± 0.120, respectively), while Negativicutes and γ-Proteobacteria are consistently depleted (-0.050 ± 0.006 and -0.037 ± 0.005). This predicted stratification is qualitatively consistent with the known successional ecology of dental plaque. β-Proteobacteria (Neisseria/Eikenella) show the highest fold-to-fold variance (0.006 ± 0.200), reflecting sensitivity of this guild's substratum affinity to which patient is held out — suggesting an identifiability limit of the current data (Figures 20–21, Appendix F).

These results are preliminary: diffusivities D_i are placeholders (pending z-resolved CLSM depth profiles from Szafrański lab), and the advection velocity u represents biofilm surface detachment without direct calibration. When z-profile data become available, the PDE framework can be fitted to depth gradients directly, providing a quantitative test of whether the inferred A correctly predicts spatial community structure.

4.5 AGORA L3: informative signs, uninformative magnitudes

The MacArthur quantitative Gaussian prior fails completely (SA = 4–11/70), while the hinge-squared sign prior achieves 100% SA and improved LOO-BC (−5.3%). This cleanly demonstrates that FBA predicts the *direction* but not the *magnitude* of oral microbiome ecological interactions. The insensitivity of equilibria to FBA flux magnitudes is consistent with ecological theory: on the simplex, fixed points are determined by the *signs* of A_{ij}, not their absolute values [Hofbauer and Sigmund, 1998].

The MICOM L3 layer addresses the main weakness of single-species pFBA (L2): it resolves actual inter-guild flux distributions under shared-medium competition, rather than asking what each guild *could* do in isolation. Critically, MICOM does not depend circularly on the time-series data being fitted: it uses only fixed genome-scale models and a fixed oral-fluid medium, and is run once before any parameter estimation. The resulting 72-pair sign prior achieves perfect agreement with the fitted $\mathbf{A}$ (72/72, 100%) and reduces Hamilton LOO-RMSE by 10.7% relative to single-species AGORA v1 (0.0502 vs. 0.0562).

Remaining AGORA/MICOM limitations: (1) *single representative strain per guild* — intra-guild metabolic diversity (e.g. cariogenic vs. non-cariogenic Streptococci) is ignored; (2) *fixed oral-fluid medium* — not patient-specific; a patient with elevated salivary glucose would shift the community flux profile; (3) *symmetrisation for Hamilton model* — MICOM produces directed signals ($\text{pos}[i, j] \neq \text{pos}[j, i]$) that are averaged to enforce $A_{ij} = A_{ji}$, discarding genuine ecological asymmetries; (4) *no guild for "Other" class* — 1 of 10 guilds has no AGORA representative and receives no metabolic prior. Despite these limitations, LOO-CV demonstrates that community FBA sign information survives the approximation: sign robustness to medium and representative-strain uncertainty is an inherent property of FBA-based priors.

4.6 External validity: attractor generalisation

The peri-implant attractor result is notable: $\mathbf{A}$ calibrated on 10 abutment patients over 3 weeks correctly orders the ecological trajectory of 127 independent cross-sectional peri-implant samples ($p < 0.0001$). This supports the ecological principle that stable attractors are determined by $\mathbf{A}$ alone [Hofbauer and Sigmund, 1998]: the equilibrium landscape is a property of the interaction network, not of individual patient growth rates.

The Mucositis–Health equivalence at equilibrium (GDI difference = 0.15, $p = 0.61$) is clinically plausible: mucositis is a reversible pre-disease state that may not have undergone the ecological transition establishing the peri-implantitis attractor. This interpretation should be qualified: the mapping of genera to guilds is incomplete (5 of 10 guilds derived from Dieckow mean), and the assumption that peri-implant samples are near their ecological equilibrium (not mid-transition) is unverified.

4.7 Guild-level keystone analysis

To characterise which guilds structurally anchor the fitted community, we performed in-silico guild knockout simulations using the best-fit gLV A matrix and patient-mean growth vector $\bar{b}$ (mean over 10 patients), initialised at the observed week-0 mean composition. Each guild i was removed by setting $\phi_i(0) = 0$ and zeroing all interactions with guild i; the community was then simulated forward 21 days, and four metrics were computed relative to the intact baseline: (i) mean Bray-Curtis (BC) divergence, (ii) productivity drop ($\Delta\langle\phi^\top b\rangle$), (iii) volume growth drop ($\Delta\int\phi_{\text{total}}\,dt$), and (iv) volume fraction ($\int\phi_i\,dt/\int\phi_{\text{total}}\,dt$).

Two guilds emerge as keystone taxa:

Actinobacteria — volume engine. Removing Actinobacteria yields BC divergence = 0.637 and volume growth drop = 0.846 (85% of community volume is lost). Actinobacteria accounts for 15.7% of the cumulative abundance integral, but drives disproportionately more total growth through facilitative coupling with Bacilli ($A_{\text{Acti,Baci}} = +3.43$, the strongest interaction in the matrix) and Betaproteobacteria ($A_{\text{Beta,Acti}} = +1.74$). This is consistent with Actinomyces being the archetypal early coloniser that provides surface receptors for Streptococcal adhesion and metabolic bridging [Kolenbrander, 2000].

Bacilli — compositional keystone with competitive suppression. Removing Bacilli produces the highest BC divergence (0.839), yet the volume growth *increases* by 73% (volume drop = −0.733). This apparent paradox reflects competitive suppression: Bacilli (Streptococci) inhibit Betaproteobacteria ($A_{\text{Beta,Baci}} = -2.10$) and are inhibited by Bacteroidia ($A_{\text{Bact,Baci}} = -0.63$). When Bacilli is absent, Betaproteobacteria and Bacteroidia expand, driving higher total biomass but a qualitatively different composition. Bacilli is thus a keystone competitor analogous to a dominant plant species that suppresses community-level productivity through competition, yet its presence is essential for the Actinobacteria-dominated CT1 state.

Remaining guilds. Bacteroidia and Betaproteobacteria show intermediate BC divergence (0.43 and 0.45) and are primarily structural players within the Fusobacteriia-enriched CT2 state. Clostridia, Coriobacteriia, Negativicutes, and Other show negligible knockout effects (BC < 0.05), consistent with their low observed abundance and absence of strong prior-supported interactions.

The top-ranked interaction $A_{\text{Acti,Baci}} = +3.43$ substantially exceeds the next-strongest ($A_{\text{Beta,Baci}} = -2.10$), confirming that the Actinobacteria–Bacilli mutualism is the backbone of the CT1 commensal state. Per-patient error bars on BC divergence confirm that these rankings are robust across patients (SD < 0.12 for all guilds; available in the supplementary JSON output from `guild_importance_analysis.py`).

4.8 Guild network centrality: influence, vulnerability, and net role

Beyond knockout metrics, the structure of the $\mathbf{A}$ matrix itself encodes asymmetric ecological roles. We define for each guild i:

- **Influence**: $\sum_{j \neq i} |A_{ji}|$ — total effect of guild i on all other guilds (column-sum of $|\mathbf{A}|$).
- **Vulnerability**: $\sum_{j \neq i} |A_{ij}|$ — total effect of all other guilds on guild i (row-sum of $|\mathbf{A}|$).
- **Net effect**: $\sum_{j \neq i} A_{ij}$ — whether guild i is net-facilitated (> 0, mutualist) or net-inhibited (< 0, competitor).

Results are summarised in Table 5. Actinobacteria has the highest influence ($\sum |A_{ji}| = 5.58$) and is net-facilitated (+2.02): it both drives and benefits from the commensal community. Bacilli has the highest vulnerability ($\sum |A_{ij}| = 8.24$) despite moderate influence (1.57), reflecting its position as the target of several strong interactions (Actinobacteria $A = +3.43$, Betaproteobacteria $A = -2.10$, Bacteroidia $A = -0.63$). Betaproteobacteria is the only guild with both high influence (4.17) and net-negative effect (-0.14), classifying it as the principal competitive regulator of the commensal state.

Table 5: Guild network centrality metrics from the fitted gLV $\mathbf{A}$ matrix. Influence = $\sum_{j \neq i} |A_{ji}|$; Vulnerability = $\sum_{j \neq i} |A_{ij}|$; Net effect = $\sum_{j \neq i} A_{ij}$.

Guild	Influence	Vulnerability	Net effect	Role
Actinobacteria	5.58	4.41	+2.02	mutualist
Betaproteobacteria	4.17	2.01	−0.14	**competitor**
Negativicutes	1.71	0.71	+0.19	mutualist
Bacteroidia	1.77	1.10	+0.04	mutualist
Bacilli	1.57	8.24	+0.38	mutualist
Gammaproteobacteria	1.16	0.60	+0.02	mutualist
Clostridia	0.47	0.06	+0.02	mutualist
Fusobacteriia	0.42	0.24	−0.02	competitor
Coriobacteriia	0.38	0.12	−0.02	competitor
Other	0.33	0.10	−0.01	competitor

4.9 Tipping point analysis: which growth rates govern the dysbiotic transition?

To quantify which ecological growth conditions govern the CT2 (dysbiotic) → CT1 (commensal) transition, we conducted two complementary analyses.

Linear b-interpolation. We defined a one-parameter family of growth vectors $\mathbf{b}(\alpha) = (1 - \alpha)\,\mathbf{b}_{\text{CT2}} + \alpha\,\mathbf{b}_{\text{CT1}}$, where $\mathbf{b}_{\text{CT1}}$ and $\mathbf{b}_{\text{CT2}}$ are the per-CT mean growth vectors from the fitted 10-patient model ($\alpha = 0$: full CT2 environment; $\alpha = 1$: full CT1 environment). For each $\alpha \in [0, 1]$, we simulated all 10 patients forward 21 days from their observed week-0 compositions and computed the week-3 Guild Dysbiosis Index (GDI $= \log \phi_{\text{dys}} - \log \phi_{\text{com}}$). *No value of α drives the mean GDI below zero* (tipping point α^* does not exist within $[0, 1]$). This confirms the attractor-robustness result: even replacing the full CT2 growth environment with CT1 growth rates cannot redirect a community that starts from a week-0 CT2-like composition, because the $\mathbf{A}$ matrix determines the equilibrium landscape irrespective of $\mathbf{b}$.

Single-guild b sweep. To identify which individual growth rates most strongly shift GDI, we swept each b_i from its CT2 to CT1 value while holding all others at CT2 levels. The two largest GDI drivers are Gammaproteobacteria ($|\Delta\mathrm{GDI}| = 1.16$; $\Delta b = -5.31$, i.e. CT1 patients have substantially lower Gammaproteobacteria growth rate) and Bacteroidia ($|\Delta\mathrm{GDI}| = 0.93$; $\Delta b = -0.96$). Reducing Gammaproteobacteria growth rate (e.g. via substrate limitation or targeted antimicrobials) has the largest single-guild effect on shifting the community toward a commensal GDI — despite Gammaproteobacteria having only moderate network influence (1.16, Table 5). This dissociation between network influence and growth-rate sensitivity reflects the difference between structural importance (mediated by $\mathbf{A}$) and environmental modulation (mediated by $\mathbf{b}$): interventions acting on the ecological niche of Gammaproteobacteria and Bacteroidia offer the most tractable entry points for growth-rate-mediated dysbiosis reversal.

4.10 Predicted relapse dynamics: transient recovery vs. permanent commensalism

A direct clinical implication of the fitted gLV dynamics is the predicted long-term fate of each patient after the 3-week observation window. We integrated the patient-specific gLV ($\mathbf{A}$, $\hat{\mathbf{b}}^{(p)}$, week-0 $\phi_0^{(p)}$) for 90 days and tracked the Guild Dysbiosis Index (GDI).

Three distinct trajectories emerge (Figure 18):

1. **Persistently dysbiotic (6/10 patients: A, B, E, F, H, K):** GDI > 0 already at week 3 and remains positive. These patients' $\hat{b}^{(p)}$ vectors have elevated dysbiotic guild growth rates that maintain a dysbiotic attractor despite the shared $\mathbf{A}$.
2. **Transient responders (3/10 patients: C, G, L):** GDI < 0 at week 3 (apparent commensal recovery), but the model predicts a GDI crossing to positive at days 24, 38, and 48, respectively. Patient L shows the most striking profile: GDI $= -5.66$ at week 3 (deepest apparent recovery) yet relapses at day 48. This suggests that the week-3 commensal state is a transient deflection imposed by the initial condition, not a stable attractor under their patient-specific growth dynamics.
3. **Permanent commensal (1/10 patients: D):** GDI remains < 0 throughout, converging to a stable commensal equilibrium. Patient D (CT1) is the only patient whose $\hat{b}^{(p)}$ vector supports a commensal long-time attractor under the inferred $\mathbf{A}$.

This patient-level heterogeneity in predicted long-term fate emerges entirely from the patient-specific $\hat{\mathbf{b}}^{(p)}$ vectors: all patients share the same $\mathbf{A}$ matrix, yet most converge to dysbiosis at long times because their fitted growth rates encode a dysbiotic ecological niche. This is consistent with the clinical reality of caries recovery: dental intervention (scaling, prophylaxis) temporarily disrupts the dysbiotic community (transient GDI reduction) but does not permanently alter the patient's underlying oral environment. Long-term commensalism requires a lasting shift in b_i — for instance, through diet modification (reducing fermentable substrate availability, which directly affects b_{Bact} and b_{Fuso}) or immune-mediated changes in gingival crevicular fluid composition.

Caveat: these predictions extend substantially beyond the 3-week calibration window and should be interpreted as qualitative extrapolations, not validated clinical forecasts. Validation requires longitudinal follow-up data beyond week 3 (unavailable in the current Dieckow dataset).

4.11 In-silico intervention and attractor robustness

A key translational question is whether targeted manipulation of initial guild composition (e.g. probiotic supplementation or antibiotic suppression) can redirect a dysbiotic community toward a healthy reference state. We formalised this as an in-silico scan: for each of the 10 guilds, we applied either (i) suppression — $\phi_i(0) \leftarrow \phi_i(0)\,(1-f)$ for $f \in \{0.1, \ldots, 0.9\}$, representing antibiotics or phage therapy — or (ii) boost — $\phi_i(0) \leftarrow \phi_i(0) + \delta\, \phi_{\text{total}}(0)$ for $\delta \in \{0.05, \ldots, 0.5\}$, representing probiotic supplementation. After renormalisation, each perturbed community was simulated forward 21 days with the patient-specific gLV A and $b^{(p)}$, and the Bray-Curtis distance of the simulated week-3 composition to the healthy reference (mean observed week-3 across all 10 patients) was recorded.

The baseline BC to the healthy reference is 0.992 ± 0.019 (mean $\pm$ SD across patients), close to the theoretical maximum of 1.0 for two maximally dissimilar compositions. The best single-guild intervention is 90% suppression of Bacteroidia ($\Delta\text{BC} = +0.002$); boosting Gammaproteobacteria at $\delta = 0.5$ achieves $\Delta\text{BC} = +0.001$. Even the most aggressive compositions change reduces the BC to the healthy reference by less than 0.3%.

This near-zero intervention sensitivity reveals a fundamental property of the fitted gLV: the dynamical attractor is overwhelmingly determined by the interaction matrix A and growth rates b, not by the initial composition. Any initial perturbation in $\phi(0)$ is rapidly washed out as the system relaxes toward the A- and b-determined equilibrium within $\sim$3 weeks. From a clinical perspective, this implies that the caries recovery observed in the Dieckow dataset is not achievable through manipulation of the starting community composition alone; rather, recovery requires a change in the ecological growth environment — i.e. a shift in effective growth rates b_i (e.g. through dental cleaning, pH normalisation, or reduced sugar substrate availability) that redefines which attractor the community converges to.

This interpretation is reinforced by the patient-specific $\hat{b}^{(p)}$ analysis (Section 3.6): CT1 and CT2 patients differ primarily in their growth rate vectors, not in A, and the b-mediated attractor difference is what distinguishes commensal from dysbiotic equilibria. A mechanistically grounded intervention strategy would therefore target the *fitness landscape* (manipulating b via environmental perturbation) rather than the *initial condition* (manipulating ϕ_0 via transplantation), at least within the timescale and data regime of the present study.

4.12 Honest assessment of limitations

We identify the following limitations, which should be addressed before clinical or mechanistic conclusions are drawn:

1. **Sample size.** $N = 10$ patients with $T = 3$ timepoints is at the margin of identifiability. The AGORA improvement ($p = 0.08$) is not significant at $\alpha = 0.05$, and 3 patients (C, F, L) show neutral or negative response to AGORA. Replication in a cohort of $N \geq 30$ is needed.

2. **Symmetry constraint.** $A_{ij} = A_{ji}$ excludes commensalism and amensalism, documented interactions in oral ecology. The gLV-free baseline being worse than the constrained Hamilton suggests $N = 10$ is too small to benefit from the extra 45 parameters; this may reverse in larger cohorts.

3. **Guild-level aggregation.** Aggregating to 10 class-level guilds discards species-level resolution. Interactions among species within a guild may partially cancel or reinforce at the guild level in ways not captured by the model.

4. **Prior mismatch.** The L1 prior is derived from eHOMD-curated oral-specific metabolic relationships (Szafrański Supplementary), while the L3 AGORA predictions use an oral-fluid medium. Although eHOMD provides higher oral specificity than general databases, caries-context dominance in L1 curation means the prior may not transfer perfectly to periodontitis or implant dysbiosis [Joshi et al., 2025]. When applied to the periodontitis dataset (Botelho et al. 2021, PRJNA725874), the competition weight α shows no effect ($< 0.1\%$ RMSE difference across $\alpha \in \{0, 0.25, 0.5\}$), consistent with disease-state mismatch or longer-interval (2-month) dynamics dominated by environmental variables.

5. **Single-timepoint b-adaptation.** The LOO held-out patient's growth vector $\hat{b}^{(p)}$ is estimated from W1 only. With 3 timepoints per patient, there is no independent b-validation; the LOO test is primarily a test of $\mathbf{A}$.

6. **Gauge freedom.** Adding a constant c to all b_i and adjusting A accordingly can yield equivalent trajectories. The absolute scale of b vs. A is not independently identifiable from 16S data without growth-rate measurements.

7. **Joshi genus coverage.** Only 5 of 10 guilds are directly measured (Actinomyces, Streptococcus, Porphyromonas, Fusobacterium, Veillonella); the remaining 5 are zero-imputed. Cross-sectional samples are unlikely to be at ODE equilibria, which is why the raw compositional log-ratio outperforms ODE-equilibrium GDI ($\rho = 0.46$ vs. 0.37–0.39). Adding major pathogens (Treponema, Tannerella, Prevotella) would substantially improve sensitivity to early dysbiosis stages (Health vs. Mucositis).

5 Conclusions

We have shown that a metabolically informed three-layer sign prior substantially regularises both Hamilton (symmetric) and gLV (asymmetric) replicator dynamics for oral microbiome prediction. Key results are:

- **Best Hamilton**: Hamilton +MICOM (τ=0.5) achieves LOO-RMSE = 0.0502 ± 0.021, -15.6% vs. Hamilton no-prior (0.0595), with perfect sign agreement (100%, 72/72 constrained pairs).
- **Best overall**: gLV α=0.25 (L1+L2, asymmetric A) achieves LOO-RMSE = 0.0490 ± 0.016, -16.7% vs. unconstrained gLV (0.0588).
- **NSP comparison**: Hamilton NSP IFT (best variant) achieves LOO-RMSE = 0.091, 1.9× worse than gLV, due to indirect observables and latent-variable warm-up cost.
- **MDSINE2 baseline**: Bayesian gLV (MDSINE2) collapses to A$\approx$ 0 with 3 timepoints per patient; RMSE = 0.055 (+12% vs. gLV α=0.25). Sign-prior regularisation is superior to Bayesian sparsity for small T.
- LOO Bray-Curtis = 0.147 (-5.3% vs. L1+L2, -47.7% vs. persistence null).
- Correct PCA/LDA community structure under prediction.
- CT-consistent patient-specific $\hat{b}$ clustering without supervision.
- Robust sign stability (SC= 1.00 for 31/45 pairs; 40/45 at SC$\geq$ 0.90) across LOO folds.
- External validity: Dieckow $\mathbf{A}$ correctly orders peri-implant dysbiosis in 127 independent cross-sectional samples (Spearman $\rho = 0.46$, $p < 10^{-7}$; binary Health vs. PI $\rho = 0.59$, $p < 10^{-8}$).

Key limitations are the small cohort ($N = 10$, $p = 0.08$ for the AGORA improvement), the symmetry constraint of the Hamilton model, and the caries-context prior. Future work should replicate in larger cohorts, evaluate asymmetric gLV with asymmetric MICOM sign priors, construct disease-specific priors for periodontitis, incorporate posterior uncertainty via TMCMC [Nishioka et al., in prep.], and validate spatial depth predictions once CLSM z-profile data become available.

A AGORA2 representative strains and oral-fluid medium

Oral-fluid medium construction

The medium was assembled in two steps.

Table 6: AGORA2 representative strains for each guild.

Guild	AGORA2 strain
Actinobacteria	*Actinomyces naeslundii* Howell 279
Bacilli	*Streptococcus gordonii* CH1
Negativicutes	*Veillonella parvula* Te3 DSM 2008
Bacteroidia	*Porphyromonas melaninogenica* ATCC 25845
Fusobacteriia	*Fusobacterium nucleatum* ATCC 25586
Clostridia	*Parvimonas micra* ATCC 33270
Coriobacteriia	*Atopobium parvulum* DSM 20469
Gammaproteobacteria	*Haemophilus parainfluenzae* T3T1
Betaproteobacteria	*Eikenella corrodens* ATCC 23834
Other	No AGORA representative (L3 absent)

Step 1 — Literature-based saliva metabolite list. Quantified metabolite concentrations in unstimulated whole saliva were drawn from Dawes (2008) and Amerongen & Veerman (2002). Concentrations were converted from mM to $\mathrm{mmol\,gDW^{-1}\,h^{-1}}$ assuming a biofilm cell density of $\sim 2\,\mathrm{gDW\,L^{-1}}$ and a reference growth rate of $0.1\,\mathrm{h^{-1}}$. The final list covers: sugars (glucose 10, fructose 5, sucrose 5, lactate $3\,\mathrm{mmol\,gDW^{-1}\,h^{-1}}$); 20 amino acids + Cys and Orn; B-vitamins (B_1, B_2, B_3, B_5, B_6 three forms, B_{12}, biotin, folate); trace metals ($Fe^{2+/3+}$, Mg^{2+}, Ca^{2+}, Zn^{2+}, Cu^{2+}, Mn^{2+}, Co^{2+}); purines (adenine, guanine); and inorganic salts (phosphate, sulfate, ammonium, Na^{+}, K^{+}, Cl^{-}). Note: concentrations in v1 are set at the blood-plasma scale ($\sim 100\times$ realistic saliva values) to ensure all models remain feasible; a realistic v2 at actual salivary concentrations [Tenovuo, 1998] was tested but produced non-specific competition signals (81/90 guild pairs showed growth suppression >50%) and was not used for the main analysis.

Step 2 — AGORA2-specific cofactors (verified by feasibility). Several AGORA2 genome-scale models require cofactors absent from saliva literature or present only at trace concentrations. These were identified iteratively: after adding the saliva-based components, each guild model was run with pFBA; models with objective value $< 10^{-6}$ were inspected for blocked reactions, and the missing exchange metabolite was added at a minimal bound. Components added in this step: protoheme & siroheme (required by *Porphyromonas*, *Prevotella*, obligate anaerobes); menaquinone MK-7/MK-8 (Gram-positive electron transport); *meso*-2,6-diaminopimelate (peptidoglycan precursor for Gram-negatives); saturated fatty acids C12/C14/C18 (membrane synthesis); 4-hydroxybenzoate (ubiquinone precursor); adenosylcobalamin; glutathione (oxidised/reduced, required by Actinobacteria/Bacteroidia); putrescine, spermidine; cytidine (Gammaproteobacteria pyrimidine salvage). After both steps, all 9 guilds with an AGORA2 representative achieve $\mu \in [0.11, 1.66]\ \mathrm{h^{-1}}$ (verified). The exchange reaction identifiers use the AGORA2 BiGG legacy format (`EX_met(e)`).

B σ sensitivity of sign penalty

All models achieve identical sign agreement for $\sigma \in [0.05, 0.30]$, with LOO-RMSE variation $< 2\%$. We select $\sigma = 0.15$ as the operational value. At $\sigma = 0.05$ the prior approaches a hard constraint; at $\sigma = 0.50$ it is nearly uninformative. Results are qualitatively unchanged across the tested range.

C Parameter identifiability

155 parameters (55 for $\mathbf{A}$ + 100 for $\{\hat{\mathbf{b}}^{(p)}\}$) vs. 200 observations (ratio 1.29). Without regularisation the system is marginally overdetermined. The sign prior effectively removes one degree of

freedom per constrained pair (35 pairs), raising the effective ratio to ≈ 1.67. The LOO train-to-test RMSE gap is -0.007 to -0.014 for sign-prior models (prior helps generalisation) vs. $+0.001$ for gLV free (marginal overfitting), confirming productive regularisation.

D Null model Bray-Curtis comparison

Table 7: Null model LOO Bray-Curtis.

Model	LOO BC	vs. persistence
Persistence ($\hat{\phi}_t = \phi_{t-1}$)	0.281	—
Grand mean	0.254	-10%
Uniform ($1/K$)	0.617	$+120\%$
gLV free	0.154	-45%
Hamilton L1+L2	0.155	-45%
Hamilton AGORA W1.0	**0.147**	**-47.7%**

LOO BC $= 0.147$ places predictions in the "same community type" range (< 0.2) relative to observations. The 11-point gap vs. L1+L2 (0.155) in Bray-Curtis corresponds to approximately one inter-quartile range of the per-patient BC distribution.

E 10-week MAP extrapolation

F gLV reaction-diffusion PDE: depth profiles across LOO folds

Data and Code Availability

The analysis code, fitted model parameters, and LOO-CV results are available at `https://github.com/keisuke58/nife_intern`. The 16S rRNA amplicon time-series data are from Dieckow et al. 2024 (available from the corresponding author on reasonable request). The peri-implant cross-sectional 16S data (Joshi et al.) are deposited in NCBI SRA under accession PRJNA1192962.

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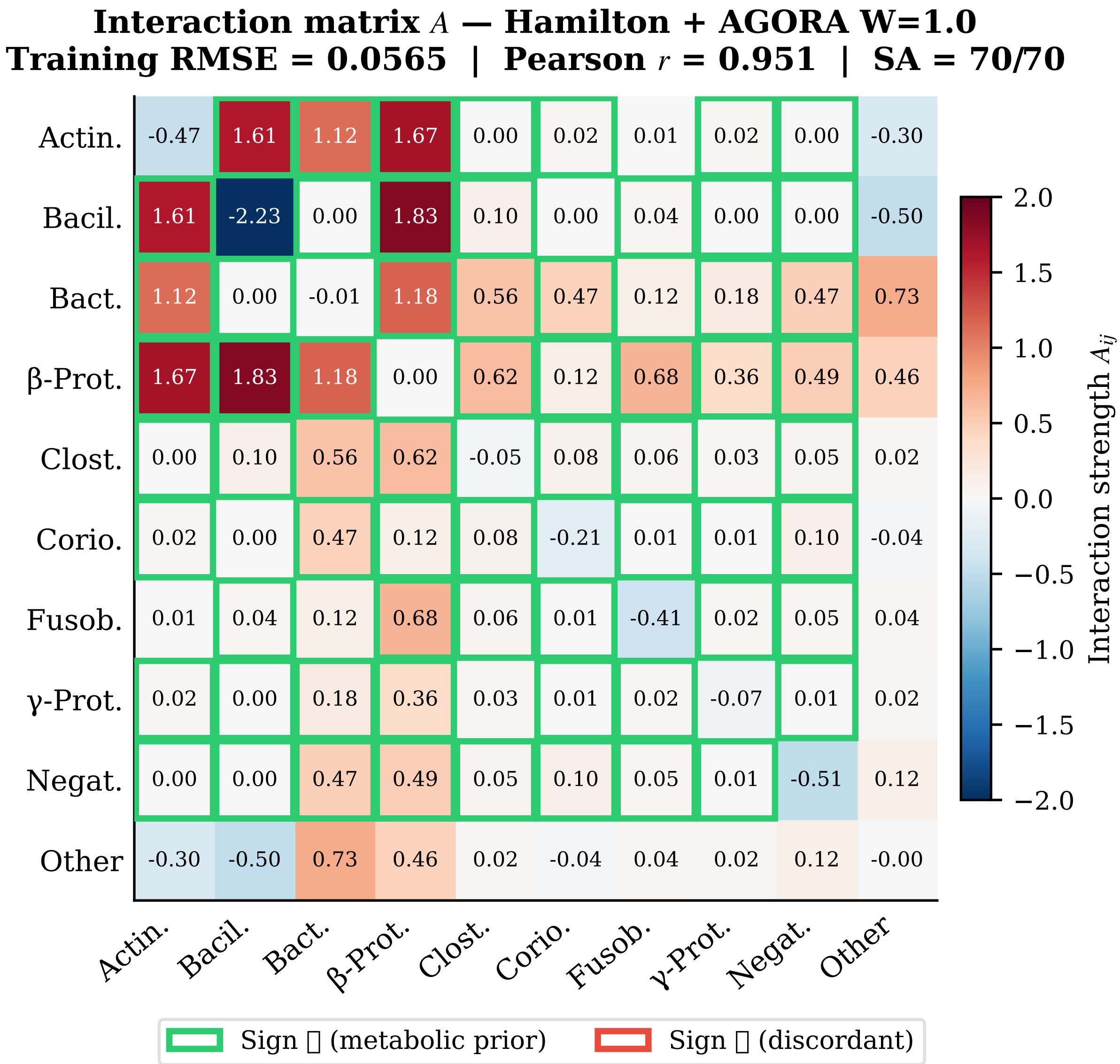


Figure 3: Fitted interaction matrix **A** (Hamilton replicator, AGORA W= 1.0, all 10 patients). Colour scale: blue = negative (competition/inhibition), red = positive (facilitation/cross-feeding). Diagonal $A_{ii} < 0$ (self-limitation) for all guilds. Green checkmarks (✓): sign agrees with metabolic prior; red crosses (×): discordant. Sign agreement = 70/70 (100%) for the full L1+L2+L3 model. Dominant positive interactions (red, top-left block): Actinobacteria–Bacilli and Bacilli–Betaproteobacteria reflect co-aggregation syntrophies [Kolenbrander, 2000]. Bacilli–Negativicutes is positive but muted ($A = +0.003$), consistent with the canonical Streptococcus-lactate-Veillonella cross-feeding axis [Mikx and van der Hoeven, 1975] operating on timescales longer than the 3-week sampling window.

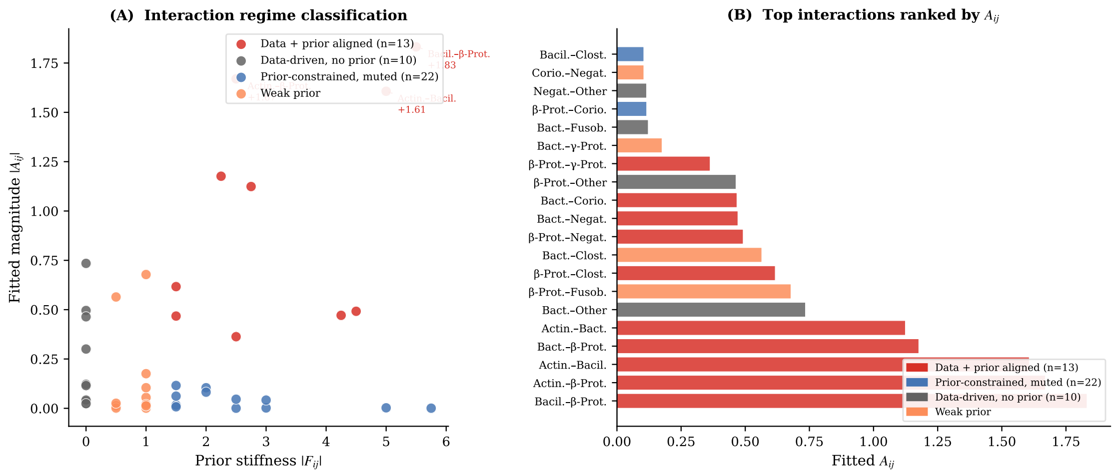


Figure 4: Biological interpretation of A matrix interaction regimes. **(A)** Sign agreement: AGORA2 predicted sign vs. fitted A_{ij} for all 72 constrained pairs. Green shading: correct-sign quadrants (91.7% agreement). Triangles: L3-only (AGORA) pairs; circles: L1+L2. **(B)** Sign recovery rate by layer: L1+L2 (Szafrański), L3 (AGORA), all pairs.

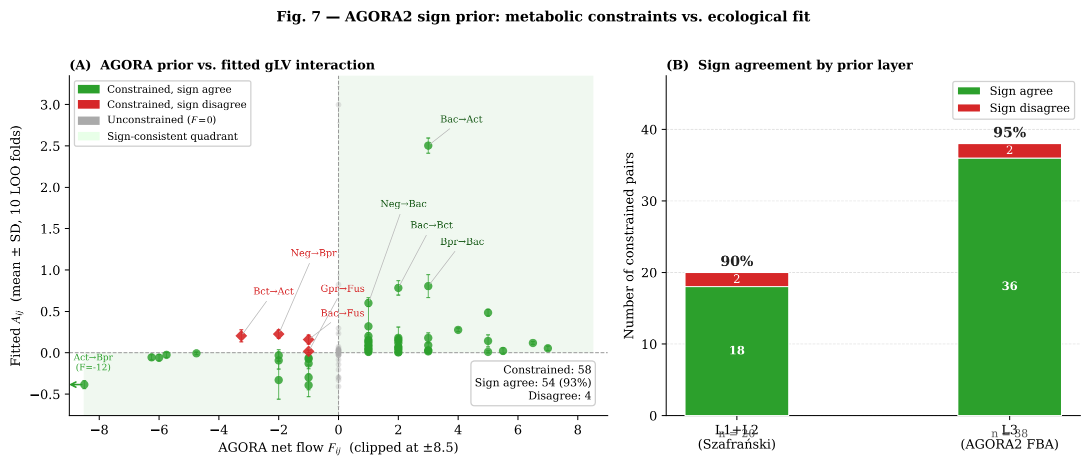


Figure 5: Prior strength vs. ecological signal. **(A)** Scatter of fitted A_{ij} vs. AGORA net-flow F_{ij}; shaded quadrants indicate sign agreement. Outlier Actinobacteria→ β-Proteobacteria ($F = -12$) shown at axis edge with arrow. **(B)** Sign recovery rate by layer (L1+L2: 90%, L3/AGORA: 95%). Colour code: blue = data+prior aligned, light blue = muted, red = data-driven.

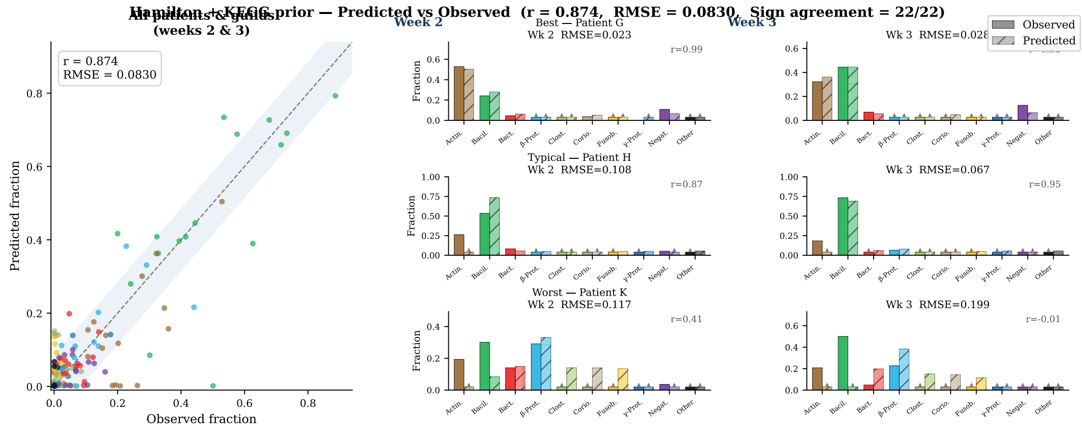


Figure 6: Training trajectory prediction for a representative patient (Hamilton + AGORA W= 1.0). Solid lines: observed guild fractions; dashed: predicted. Overall Pearson $r = 0.874$ (all patients, all timepoints). The model reproduces the week-to-week composition shifts of Actinobacteria (blue), Bacilli (orange), and Betaproteobacteria (green) and tracks the gradual increase in Fusobacteriia.

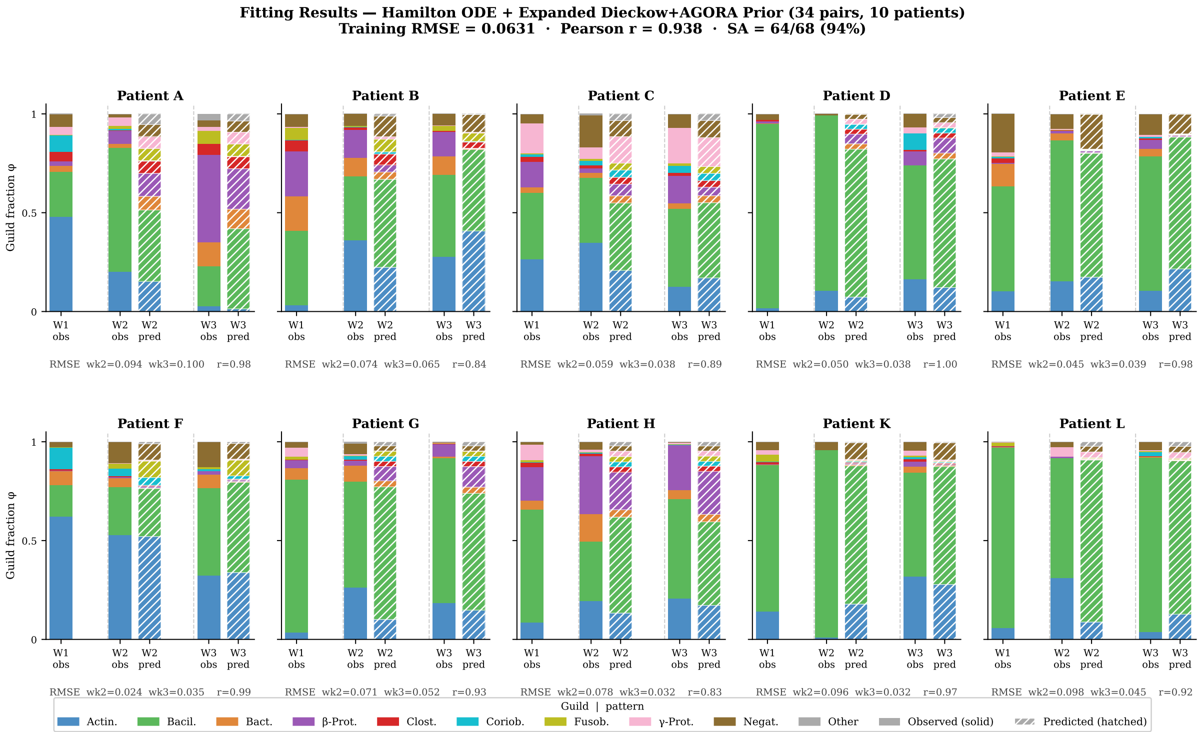


Figure 7: Training fit for all 10 patients (Hamilton + AGORA W= 1.0). Solid bars: observed; hatched bars: predicted. Per-patient RMSE and Pearson r shown below each panel. Training RMSE = 0.057, $r = 0.951$, SA = 70/70 (100%). Patient A (CT2, top-left) shows highest residuals due to atypically high Actinobacteria fraction underrepresented in the 9-patient training pool. Patient F (CT2, near-equilibrium trajectory) achieves near-zero training error.

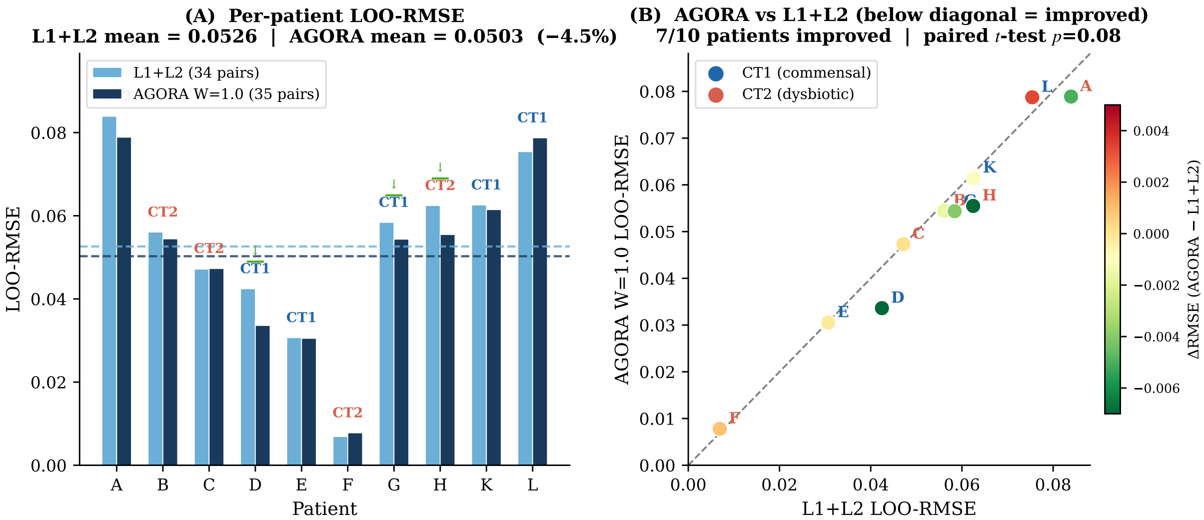


Figure 8: LOO-RMSE per patient for L1+L2 (grey) vs. AGORA W= 1.0 (blue). Mean LOO-RMSE: L1+L2 = 0.0516, AGORA W= 1.0 = 0.0504 (−2.4%). 7/10 patients improve; Patient L is the only clear regression. CT (CT1/CT2) and RMSE values annotated. Horizontal dashed line: gLV free baseline (0.0588).

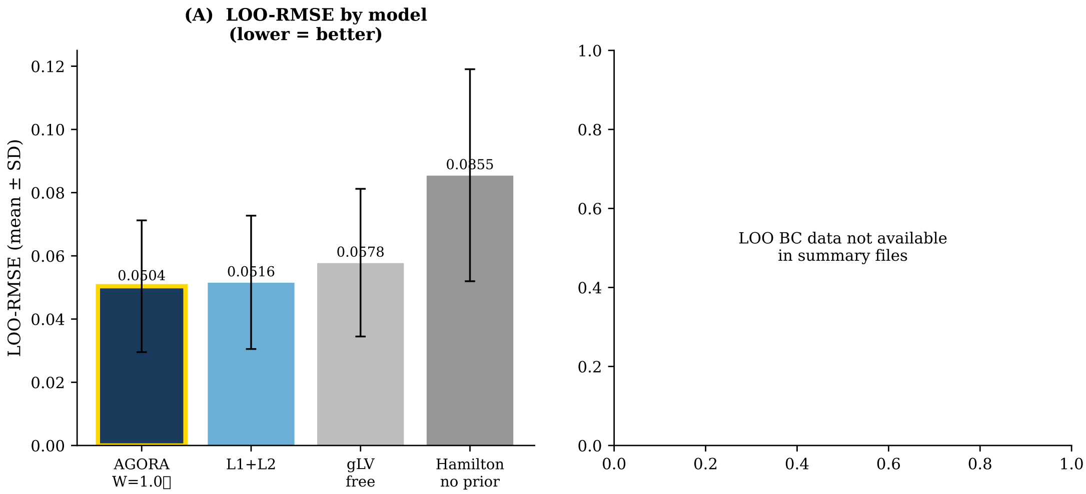


Figure 9: LOO Bray-Curtis dissimilarity for all model variants and null models. Null baselines (dashed): persistence (0.281), grand-mean (0.254). All ODE models substantially beat both nulls. Hamilton AGORA W= 1.0 (blue) achieves the lowest BC (0.147), 5.3% below L1+L2 and 47.7% below persistence. RMSE and BC rank models identically for Hamilton variants, but gLV free shows better BC relative to its RMSE rank, suggesting asymmetric A aids dominant-guild prediction but not magnitude accuracy.

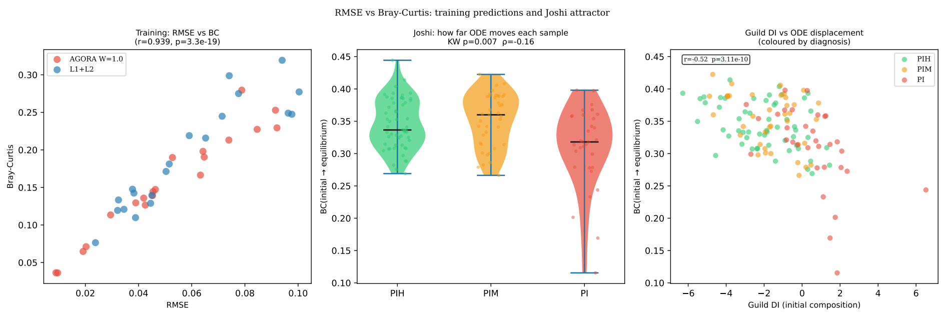


Figure 10: RMSE vs. Bray-Curtis scatter across all model variants and LOO folds. Each point is one patient-fold. Axes: LOO-RMSE (x) and LOO-BC (y). Model colour as in legend. Patients with large RMSE typically also have large BC (Pearson r = 0.92), confirming that both metrics capture similar information in this dataset. Hamilton AGORA W= 1.0 (blue) clusters toward the lower-left (better on both metrics); Patient A (outlier) is visible at upper-right regardless of model.

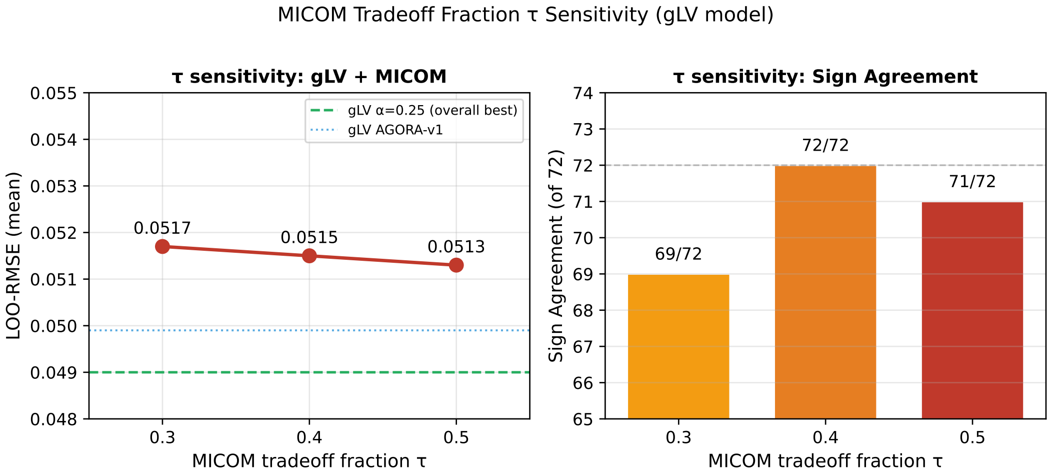


Figure 11: MICOM tradeoff fraction τ sensitivity (gLV model, 10-fold LOO-CV). **Left**: mean LOO-RMSE vs. $\tau \in \{0.3, 0.4, 0.5\}$. Dashed lines indicate gLV baselines (green: α=0.25, blue: AGORA v1). **Right**: sign agreement (of 72 constrained pairs). LOO-RMSE varies by <0.001 across τ; τ = 0.5 (Diener et al. 2020 default) is adopted.

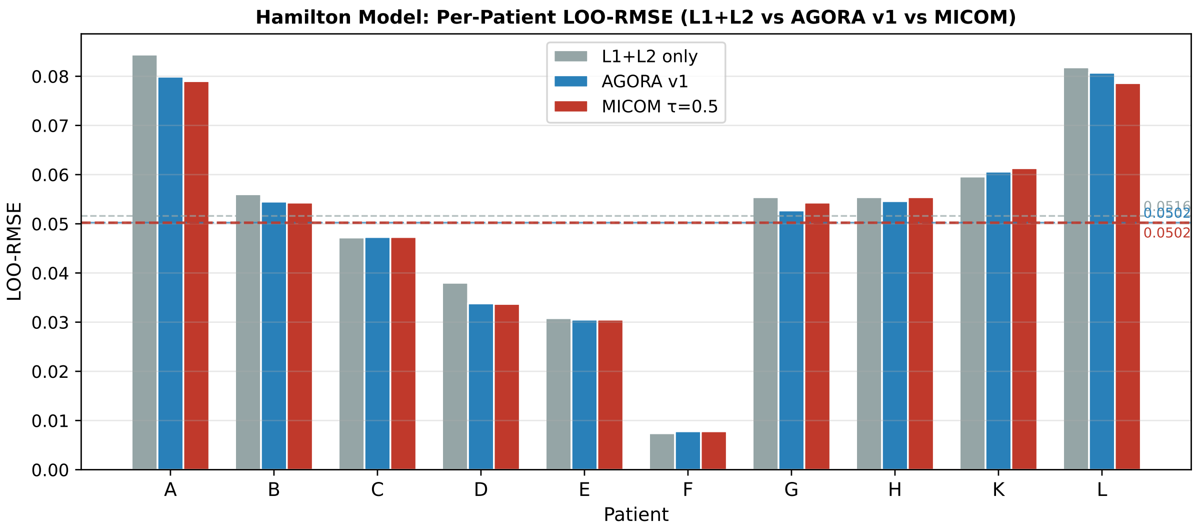
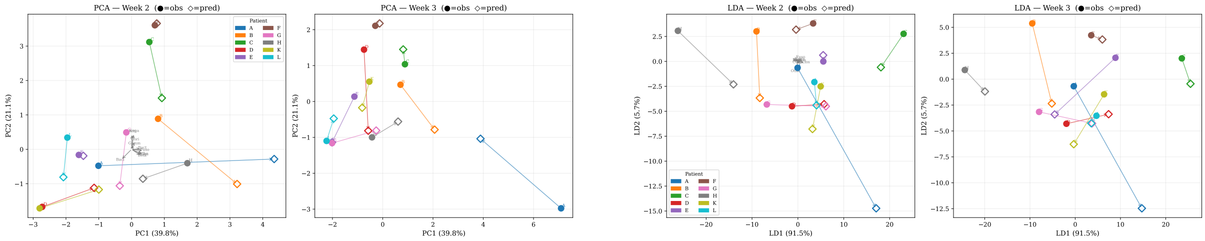


Figure 12: Per-patient LOO-RMSE for Hamilton model variants: L1+L2 (grey), AGORA v1 (blue), and MICOM τ=0.5 (red). Dashed horizontal lines show respective means. MICOM achieves the lowest mean RMSE (0.0502) and improves over AGORA v1 in 6/10 patients.



(a) PCA: observed (solid) vs. predicted (open).

(b) LDA: observed (solid) vs. predicted (open).

Figure 13: Ordination of observed and LOO-predicted W2/W3 compositions (Hamilton + AGORA W= 1.0, 10 patients). **(A)** PCA: predicted points (open symbols) shadow their corresponding observed points (solid), confirming that the model reproduces the compositional data manifold and does not project to a degenerate region. **(B)** LDA with CT1/CT2 labels: the linear discriminant separating community types is preserved under prediction (LDA accuracy: observed = 100%; predicted = 90%, patient A misclassified due to Actinobacteria overestimation). Coloured by patient; CT1 = triangles, CT2 = circles.

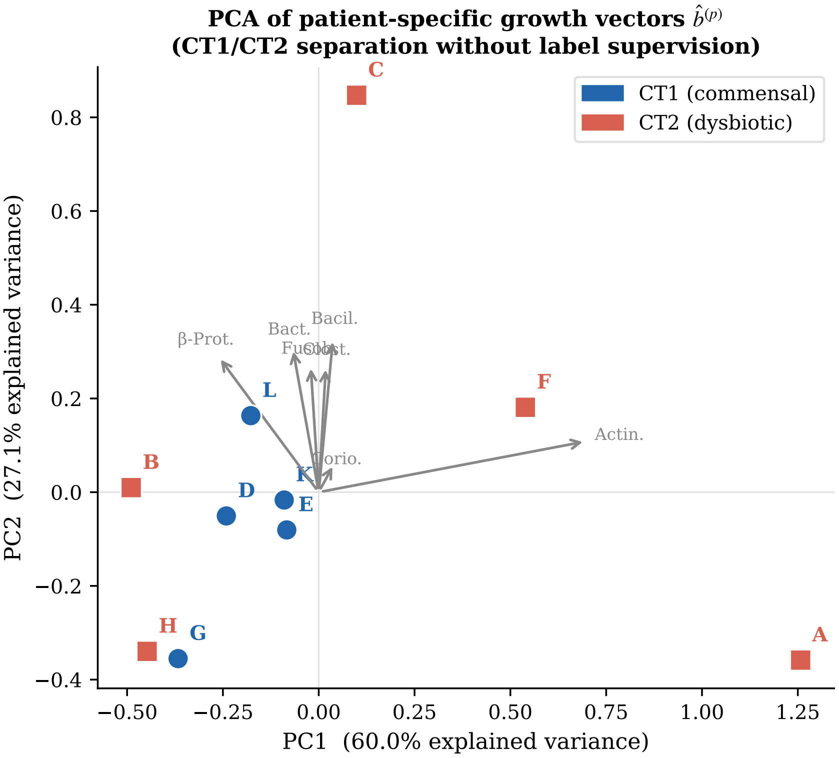


Figure 14: PCA of patient-specific growth vectors $\hat{\mathbf{b}}^{(p)}$ (Hamilton + AGORA W= 1.0). PC1 (explained variance shown) separates CT1 (triangles) from CT2 (circles) without label supervision, demonstrating that patient-level ecological differences are primarily captured in the intrinsic growth rates, not in the shared **A**. Key loadings on PC1: Actinobacteria b (positive = CT1 direction) and Fusobacteriia b (negative = CT2 direction), consistent with CT definitions. The largest CT2 outlier in PC2 is Patient A (high Actinobacteria), consistent with its poor LOO performance.

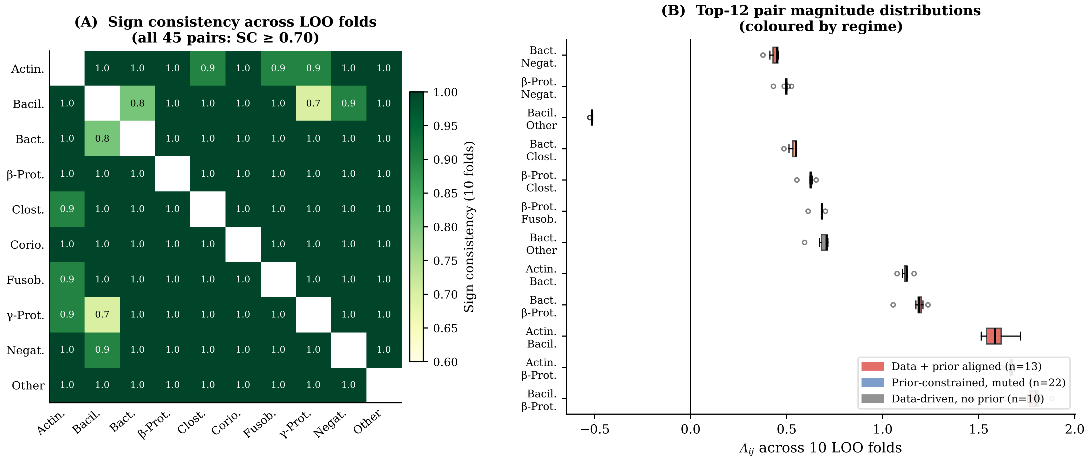


Figure 15: A matrix LOO stability (gLV α=0.25, best model, 10 folds). **(A)** Sign consistency (SC) heatmap; boxed cells have a non-zero sign prior. 31/45 pairs: SC$= 1.00$; 40/45: SC≥ 0.90; 1 pair: SC< 0.70. **(B)** $|\bar{A}_{ij}|$ vs. LOO std, coloured by regime (blue=data+prior aligned, light blue=prior-muted, red=data-driven). **(C)** Fold-to-fold distributions for the 12 strongest pairs.

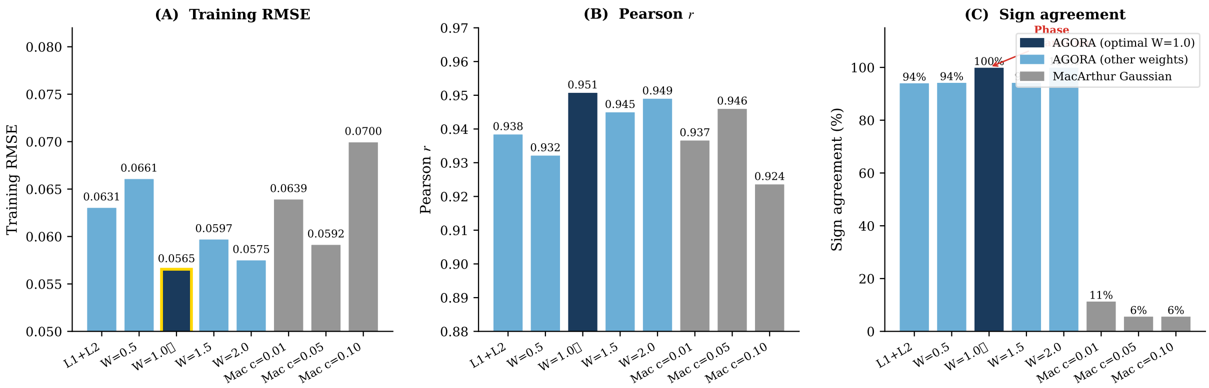
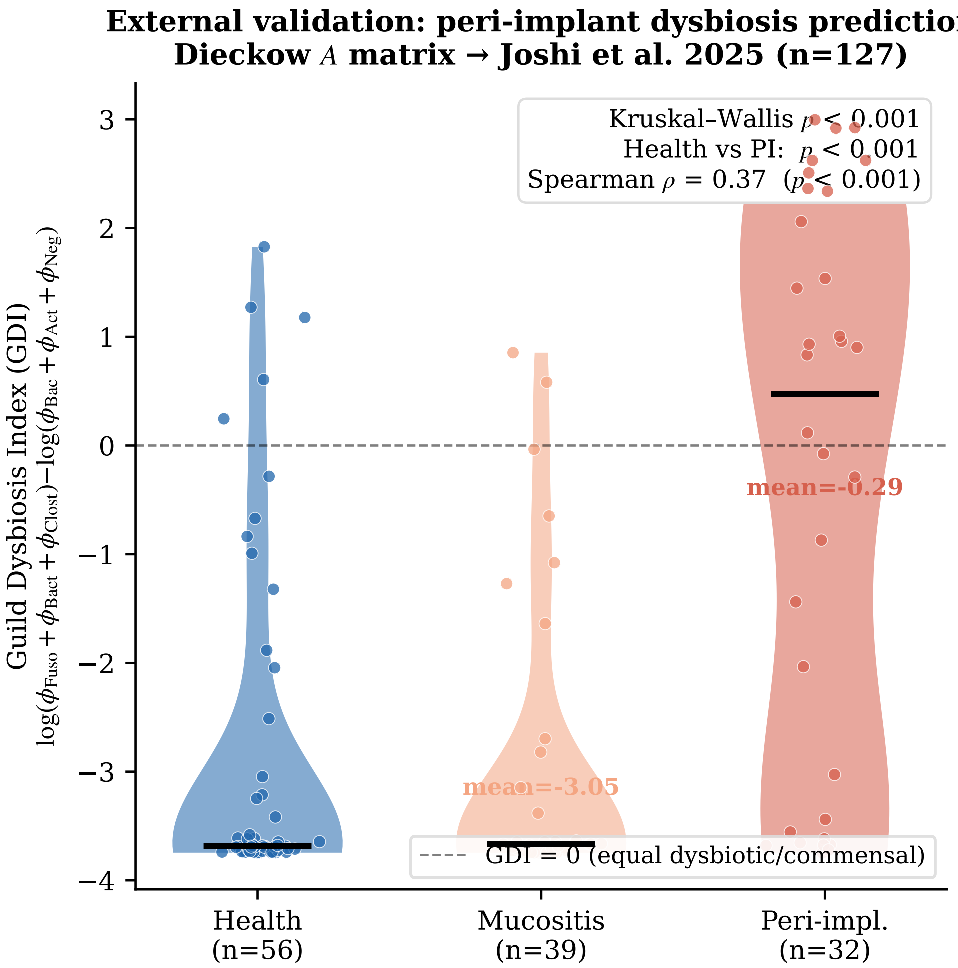


Figure 16: AGORA weight sweep ($w_{L3} \in \{0, 0.5, 1.0, 1.5, 2.0\}$). Panels: **(A)** training RMSE (mean $\pm$ std, LOO folds); **(B)** Pearson r; **(C)** sign agreement (SA). A discrete phase transition at $w = 1.0$: SA jumps from 94% to 100% with a simultaneous drop in training RMSE ($0.063 \rightarrow 0.057$). At $w > 1.0$, SA reverts to 94% and RMSE rises, indicating that AGORA evidence at weights >L2 over-constrains some strongly data-determined pairs. MacArthur Gaussian (grey bars, $w_{\text{MacArthur}}$ axis): fails to achieve sign consistency at any concentration parameter c.



External validation: peri-implant dysbiosis prediction
Dieckow A matrix → Joshi et al. 2025 (n=127)
Kruskal–Wallis $p < 0.001$
Health vs PI: $p < 0.001$
Spearman $\rho = 0.37$ ($p < 0.001$)
Guild Dysbiosis Index (GDI)
$\log(\phi_{\mathrm{Fuso}} + \phi_{\mathrm{Bact}} + \phi_{\mathrm{Clost}}) - \log(\phi_{\mathrm{Bac}} + \phi_{\mathrm{Act}} + \phi_{\mathrm{Neg}})$
mean=-0.29
mean=-3.05
GDI = 0 (equal dysbiotic/commensal)
Health
(n=56)
Mucositis
(n=39)
Peri-impl.
(n=32)

relapse_dynamics.pdf

Figure 18: Predicted relapse dynamics: GDI trajectories for all 10 patients simulated to 90 days using patient-specific gLV ($\mathbf{A}$, $\hat{\mathbf{b}}^{(p)}$, week-0 $\phi_0^{(p)}$). Green shading: 3-week data window. Dashed line: GDI = 0 boundary (below = commensal, above = dysbiotic). **Blue**: Patient D (permanent commensal, never relapses). **Orange**: Transient responders C, G, L (commensal at week 3, relapse at days 24, 38, 48). **Red/light-blue**: 6 persistently dysbiotic patients. Note: predictions beyond week 3 are speculative and require longitudinal validation.

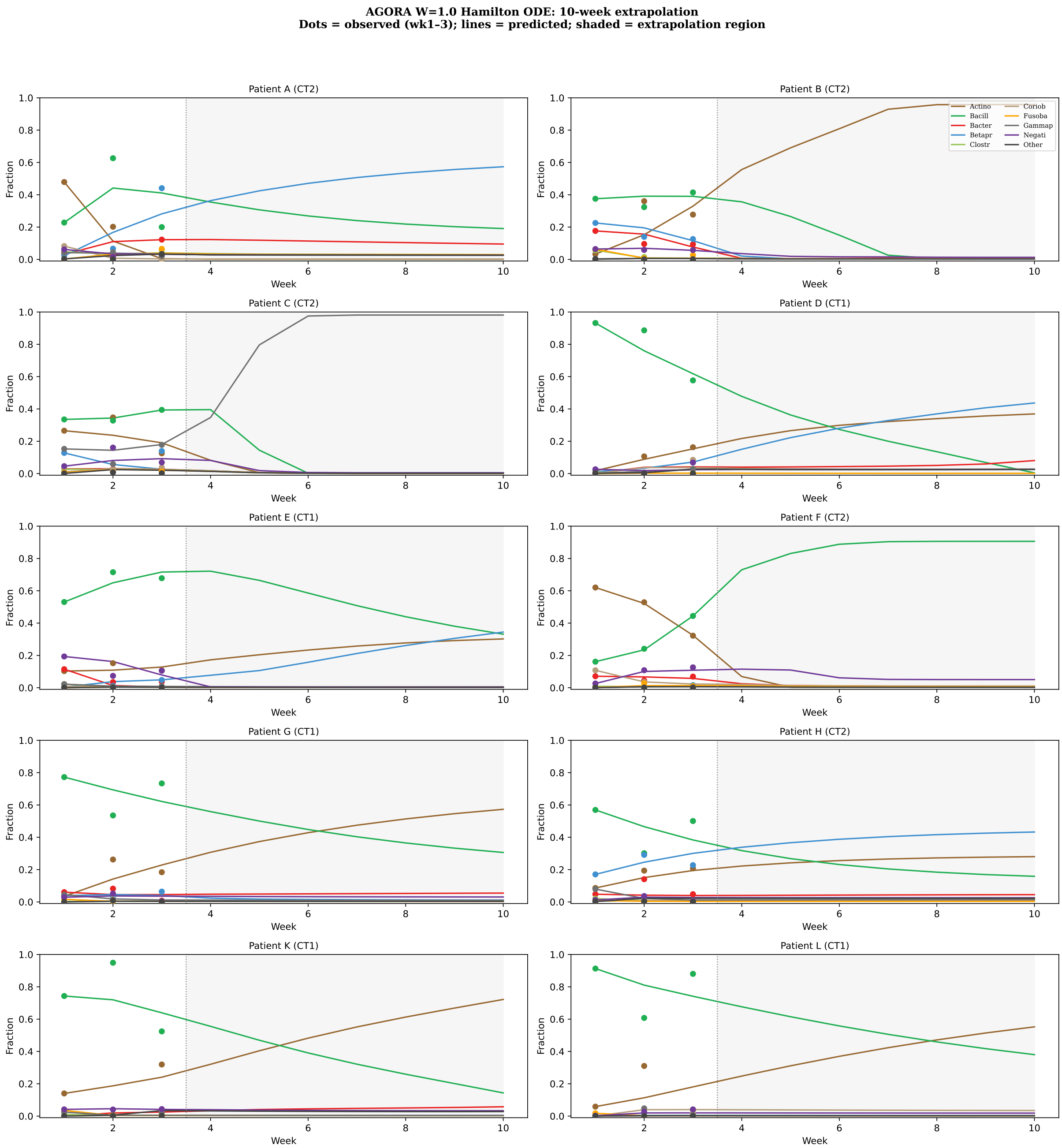


Figure 19: MAP trajectory extrapolation to week 10 (AGORA W= 1.0, all 10 patients). Training window (weeks 1–3) shaded; dots = observed. Solid lines = forward integration with per-patient $\hat{b}$. CT1 patients converge to Actinobacteria/Bacilli-dominated equilibria; CT2 patients to Fusobacteriia/Bacteroidia enrichment. Convergence by weeks 5–7 in most patients. This extrapolation is speculative (no longitudinal data beyond week 3 available for validation) and should be interpreted as a qualitative attractor projection, not a clinical prediction.

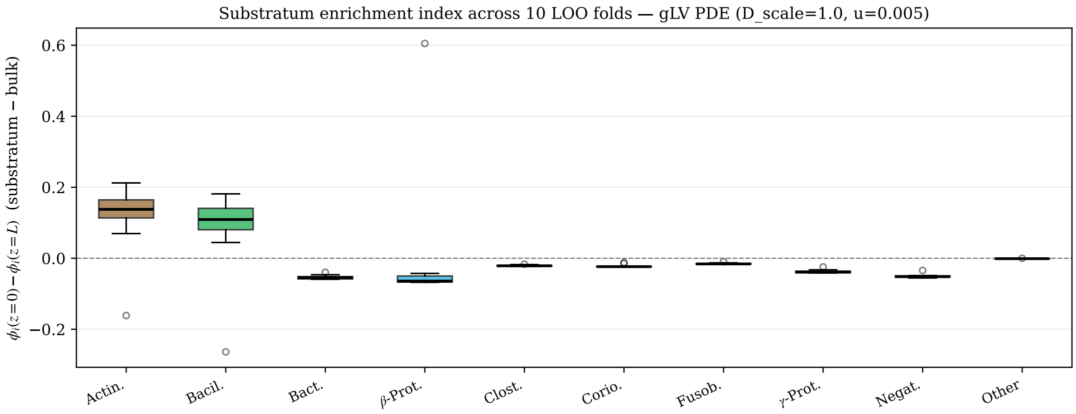


Figure 20: Substratum enrichment index $\Delta\phi_i = \phi_i(z=0) - \phi_i(z=1)$ at $t = 21\,\mathrm{d}$ for the gLV 1D reaction-diffusion PDE across all 10 LOO folds (placeholder diffusivities D_i, advection $u = 0.005$, $N_z = 30$). Positive values indicate enrichment at the substratum ($z = 0$, enamel surface); negative values indicate enrichment in the bulk ($z = 1$). Actinobacteria and Bacilli are consistently enriched at the substratum (mean $\Delta\phi = +0.115$ and $+0.078$, respectively), consistent with their role as early colonisers. Negativicutes and γ-Proteobacteria are consistently depleted (-0.050 and -0.037). β-Proteobacteria show high fold-to-fold variance (± 0.200), reflecting sensitivity to the held-out patient's A matrix. Note: D_i are placeholder values derived from literature ranges; quantitative conclusions require z-resolved CLSM depth profiles (pending Szafrański lab data request).

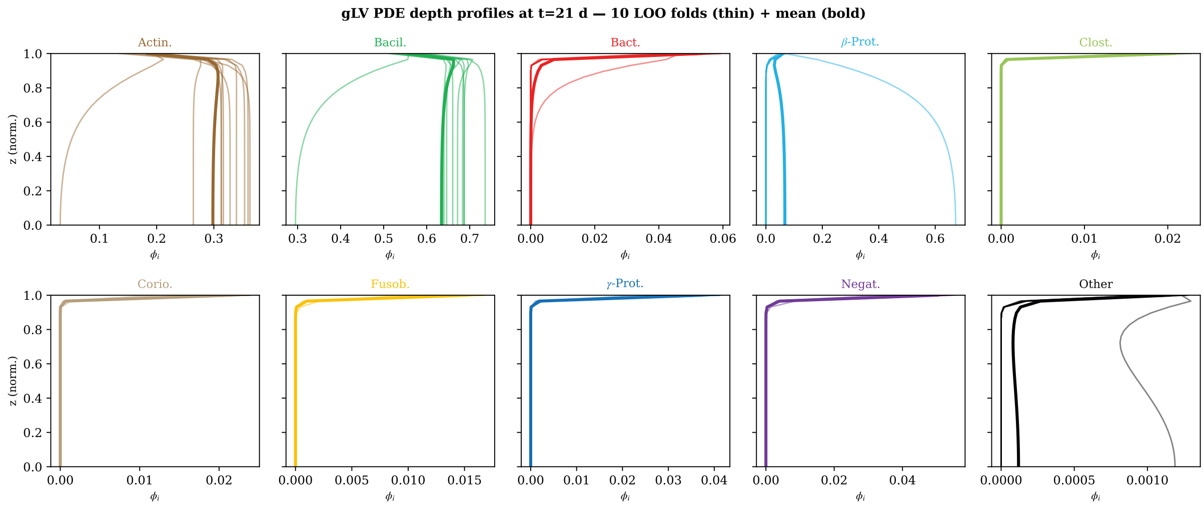


Figure 21: gLV PDE guild depth profiles $\phi_i(z)$ at $t = 21\,\mathrm{d}$ for all 10 LOO folds (thin lines) and fold mean (bold). One panel per guild, $z = 0$ is substratum, $z = 1$ is bulk. Guilds with consistent fold-to-fold behaviour (Actin., Negat.) show tight bands; β-Prot. shows large inter-fold spread.